

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

Complete

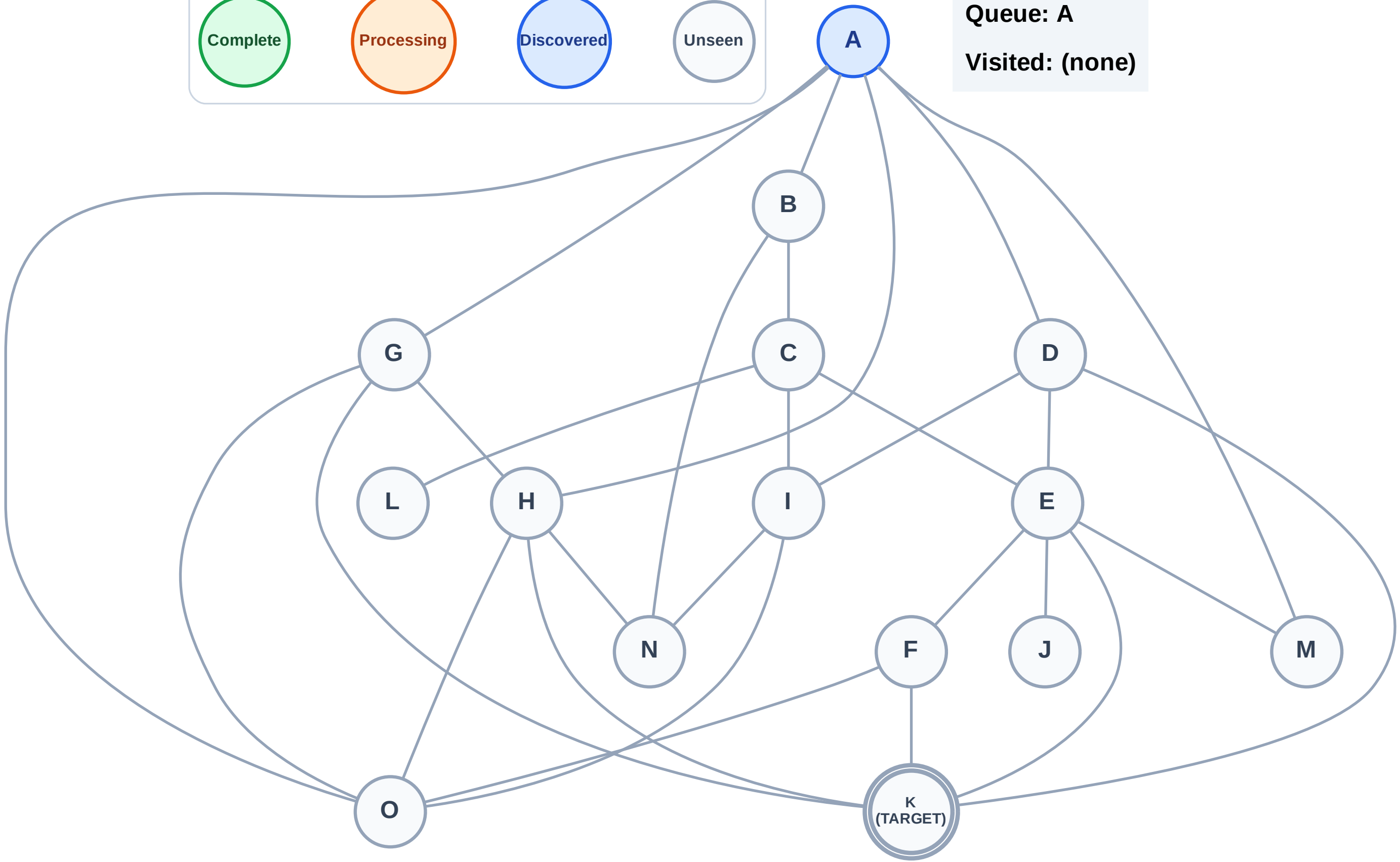
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

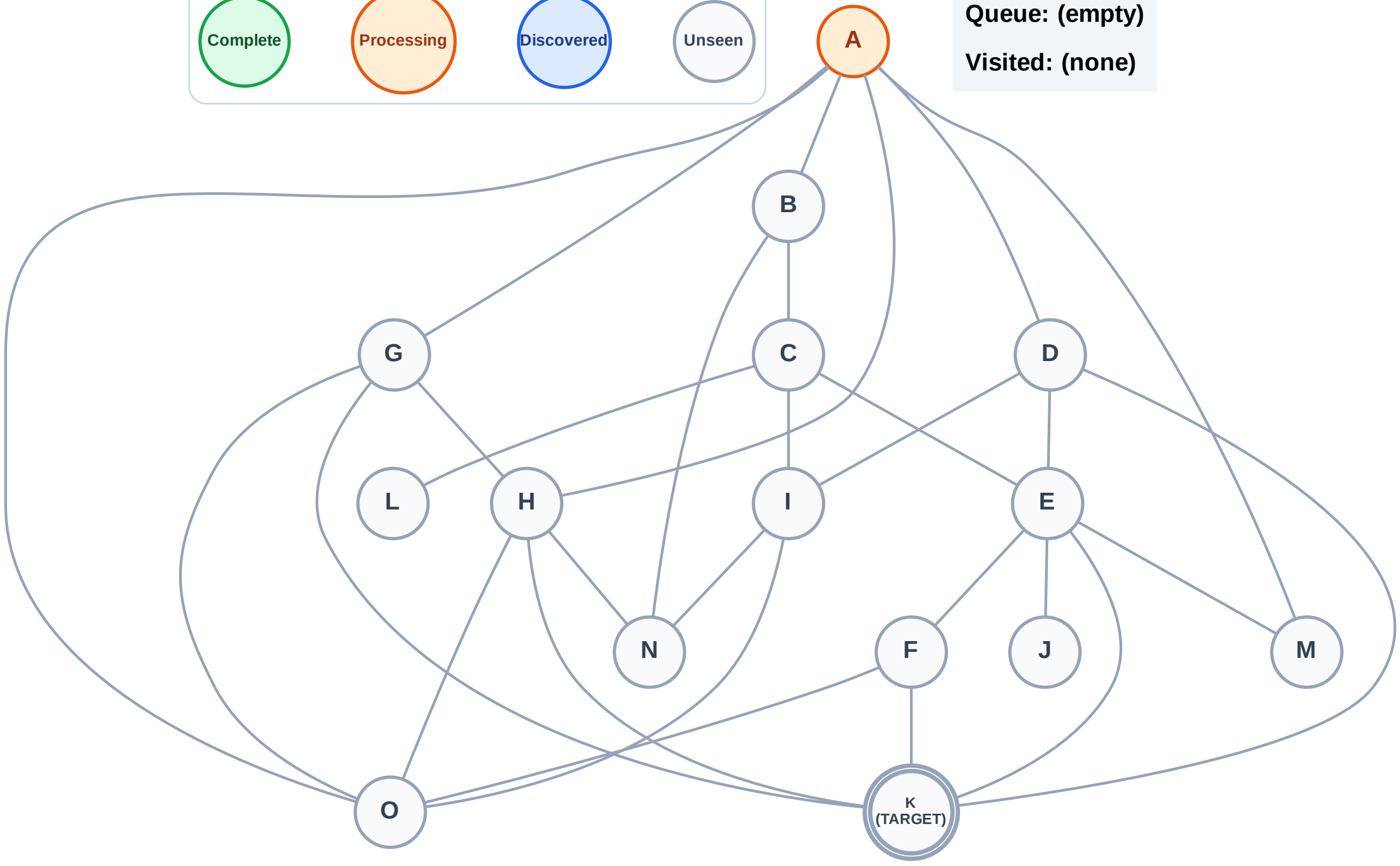
Complete

Processing

Discovered

Unseen

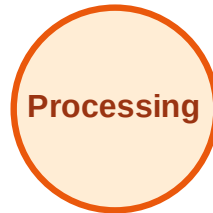
Queue: (empty)
Visited: (none)



BFS Traversal

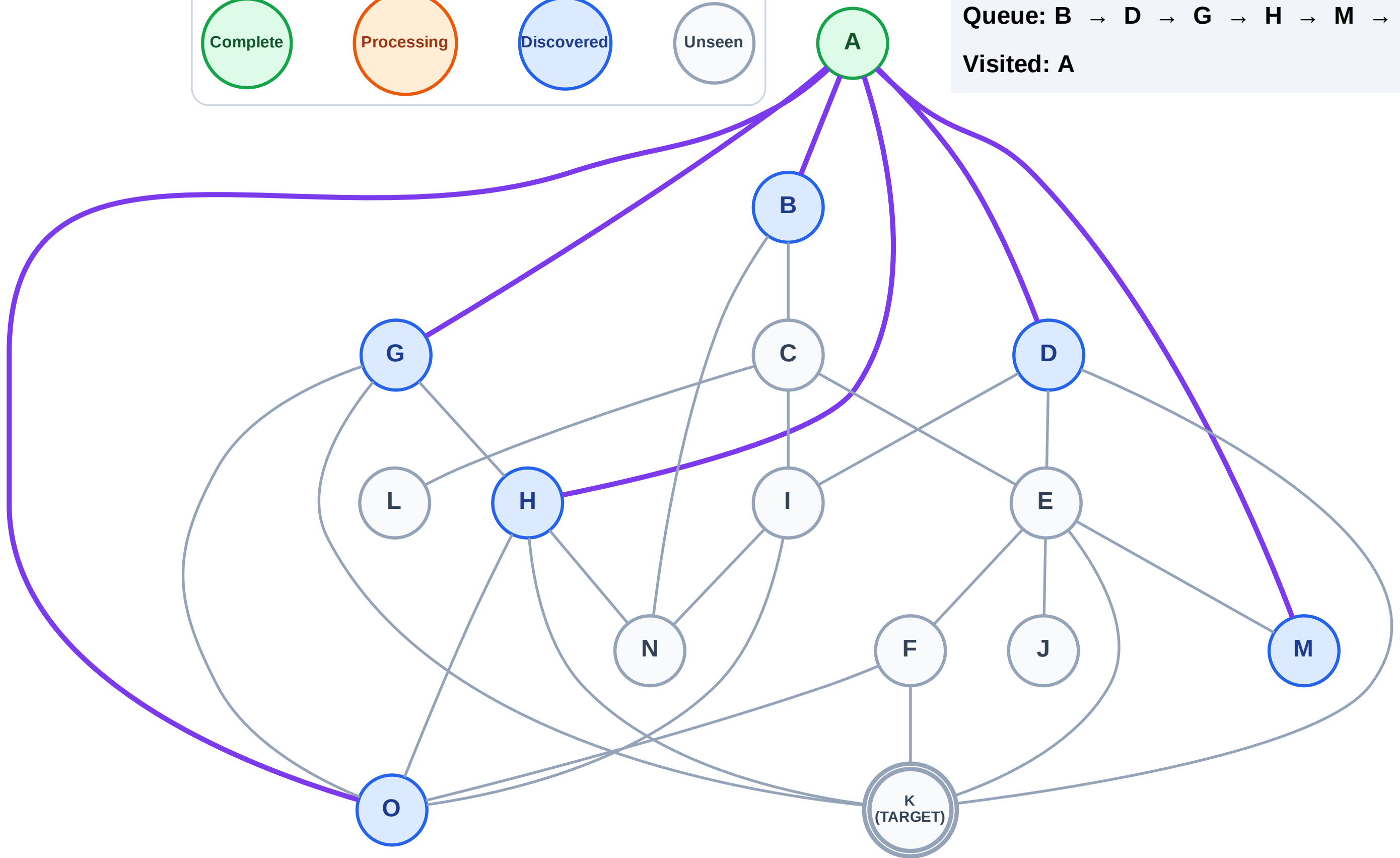
Step 3: Discover B, D, G, H, M, O from A

Legend



Queue: B → D → G → H → M → O

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

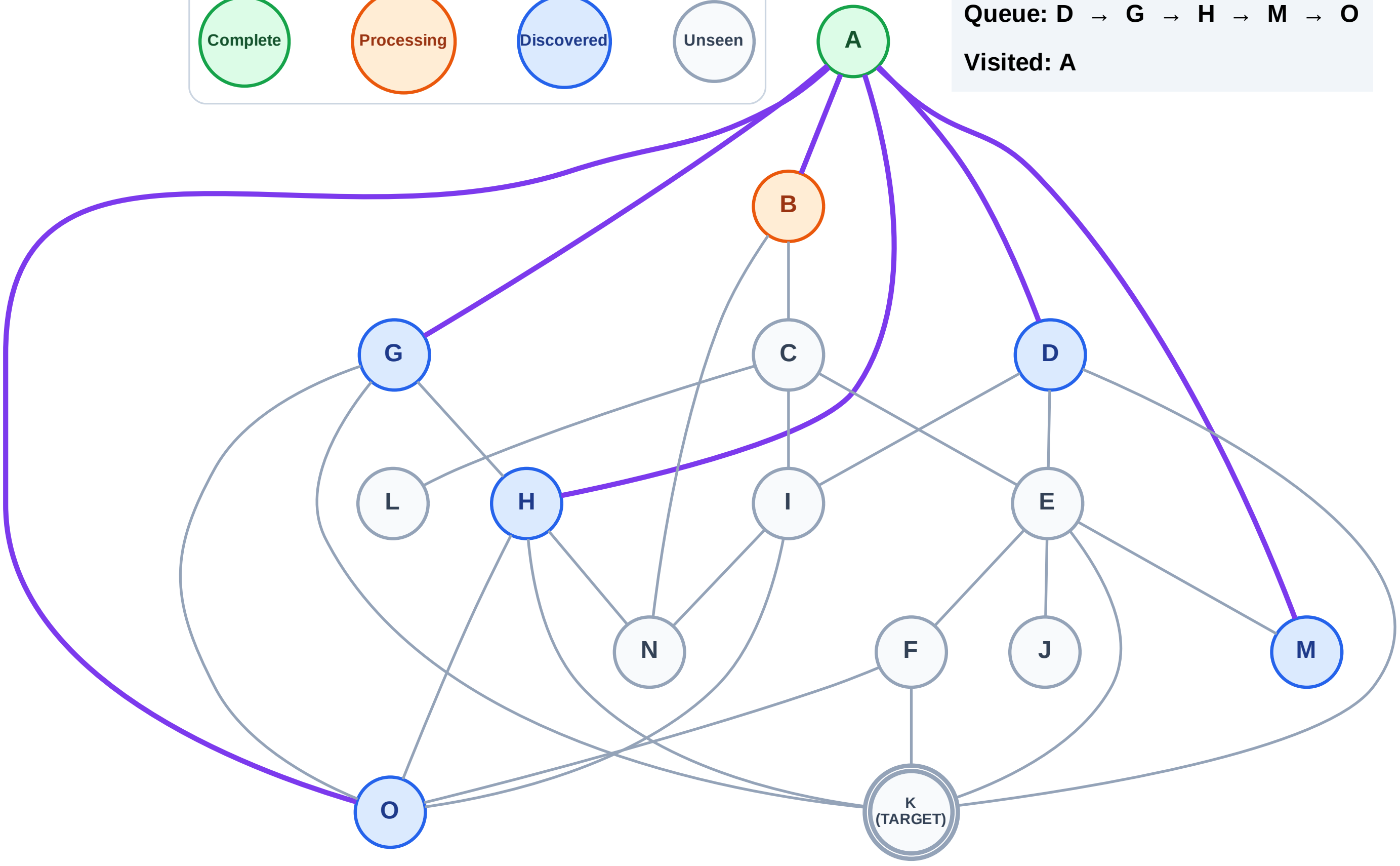
Processing

Discovered

Unseen

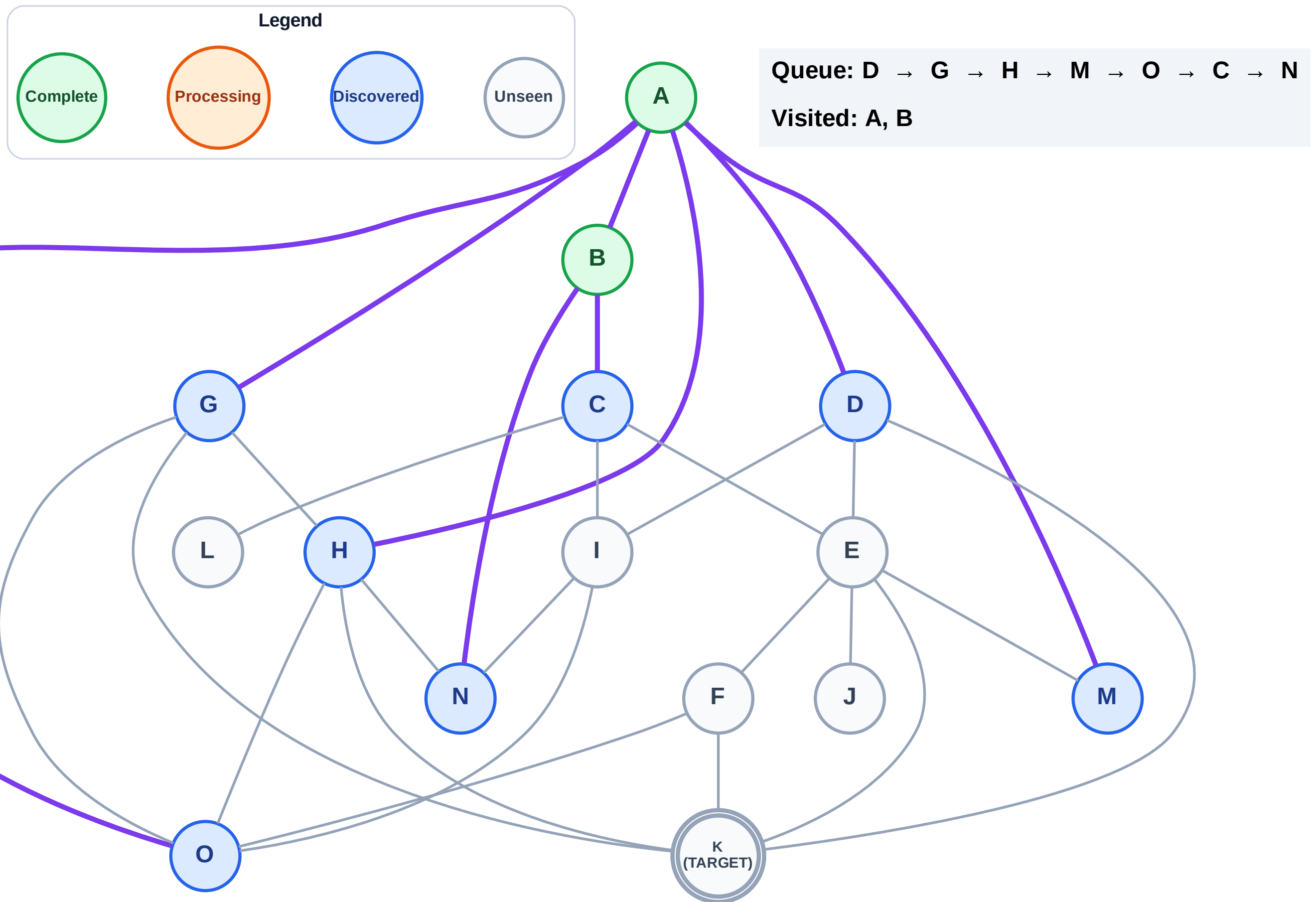
Queue: D → G → H → M → O

Visited: A



Step 5: Discover C, N from B

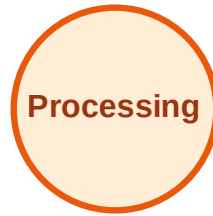
Step 5: Discover C, N from B



BFS Traversal

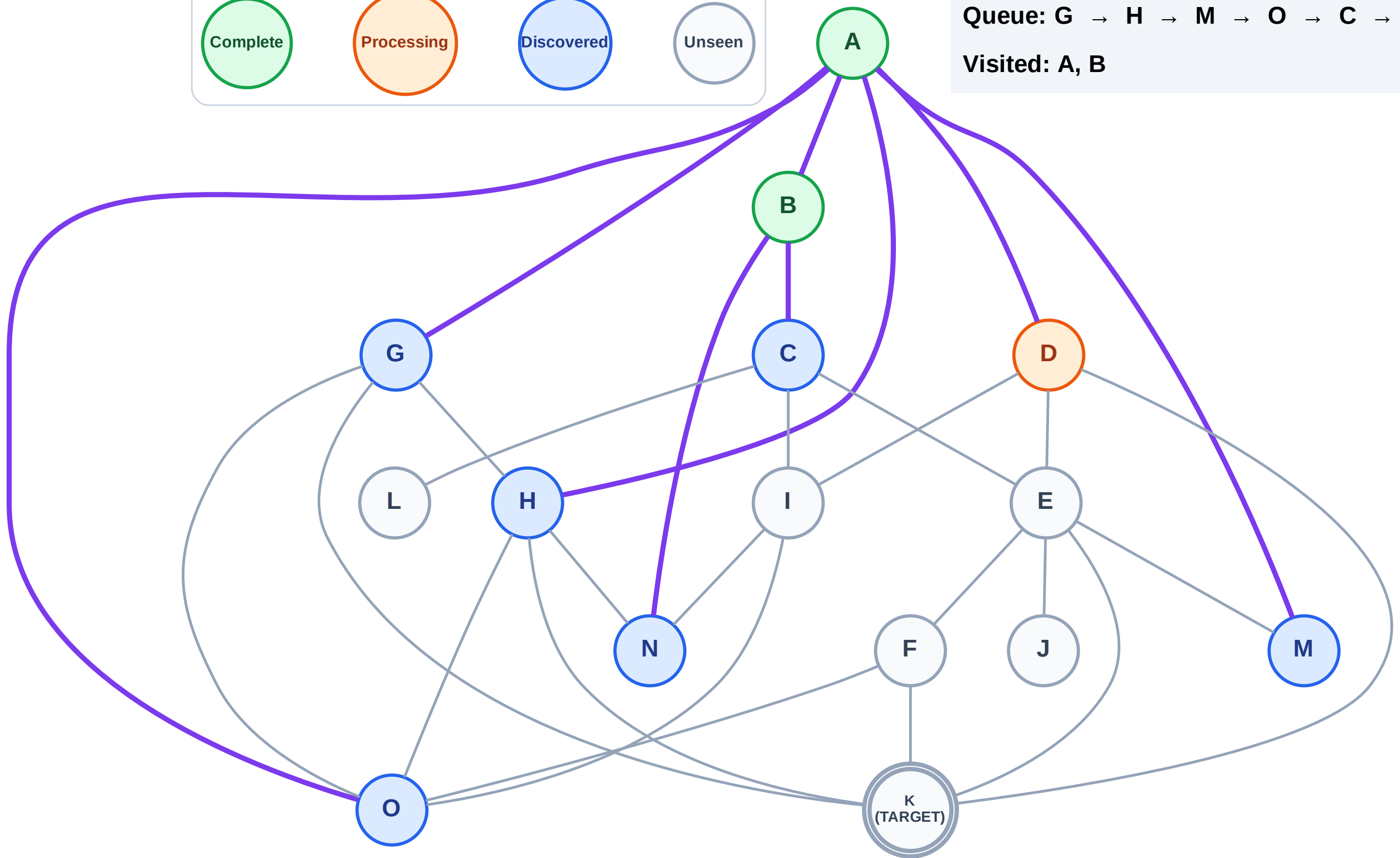
Step 6: Process node D

Legend



Queue: G → H → M → O → C → N

Visited: A, B



BFS Traversal

Step 7: Discover E, I, K from D

Legend

Complete

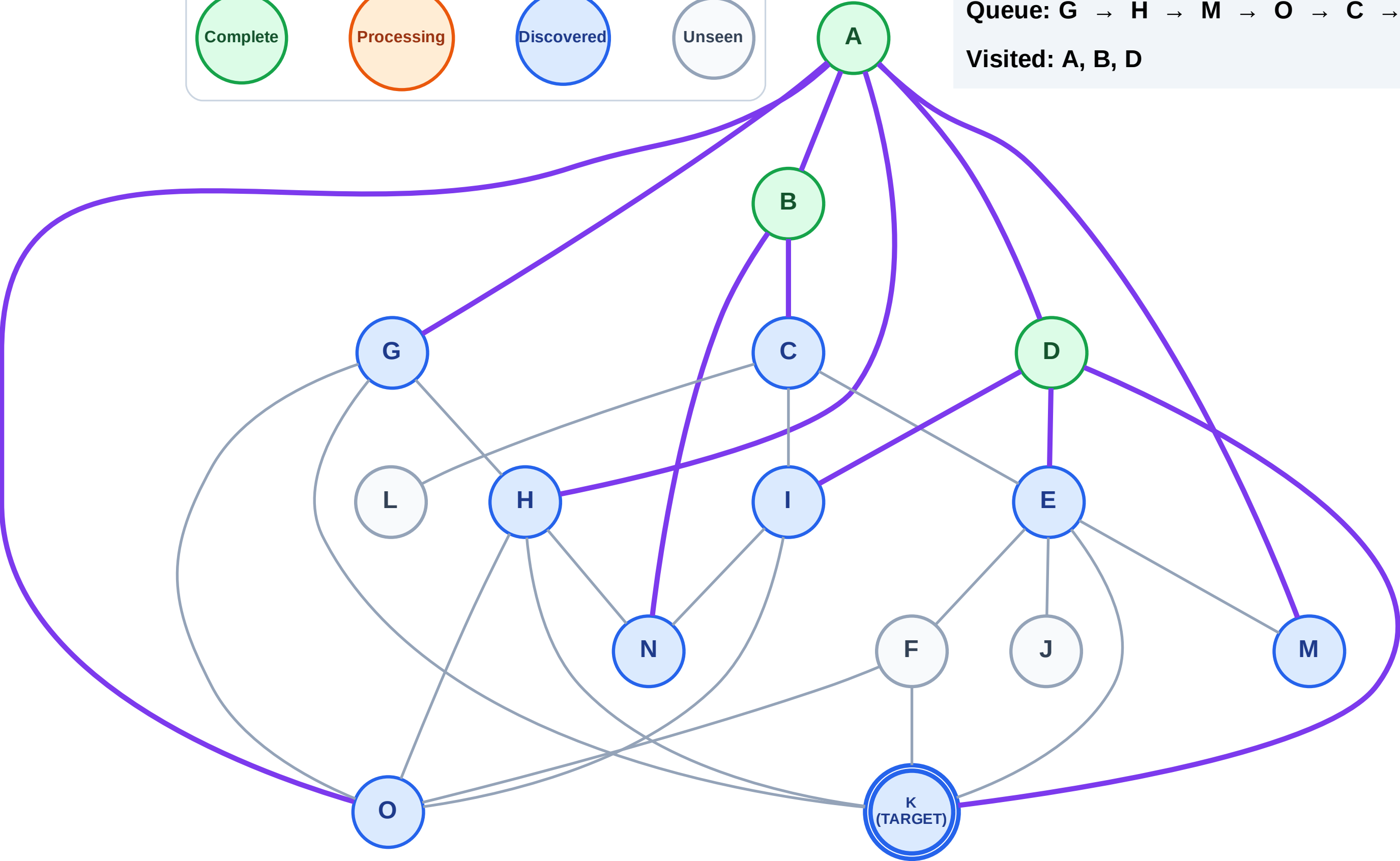
Processing

Discovered

Unseen

Queue: G → H → M → O → C → N → E → I → K

Visited: A, B, D



BFS Traversal

Step 8: Process node G

Legend

Complete

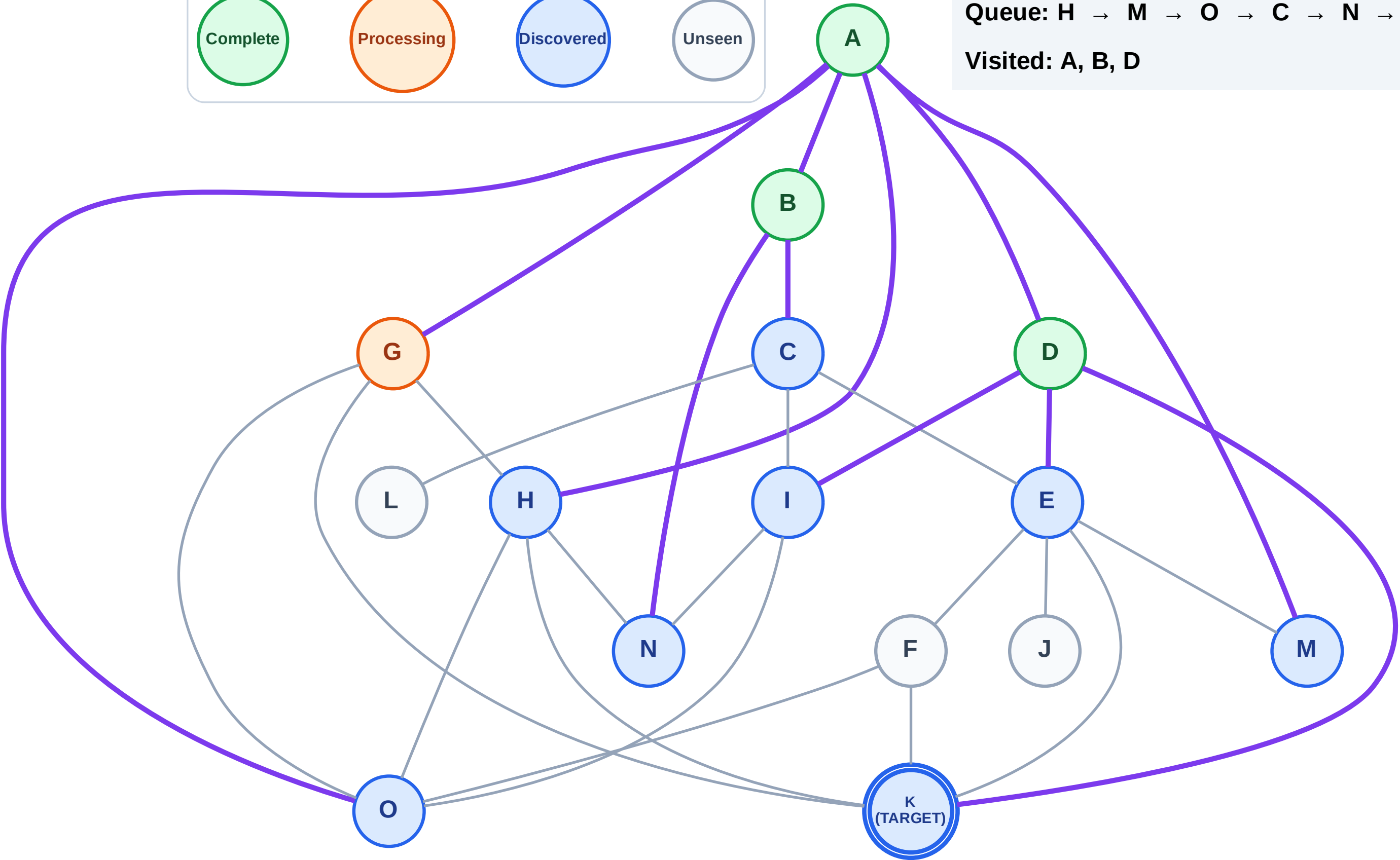
Processing

Discovered

Unseen

Queue: H → M → O → C → N → E → I → K

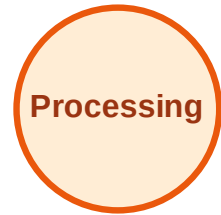
Visited: A, B, D



BFS Traversal

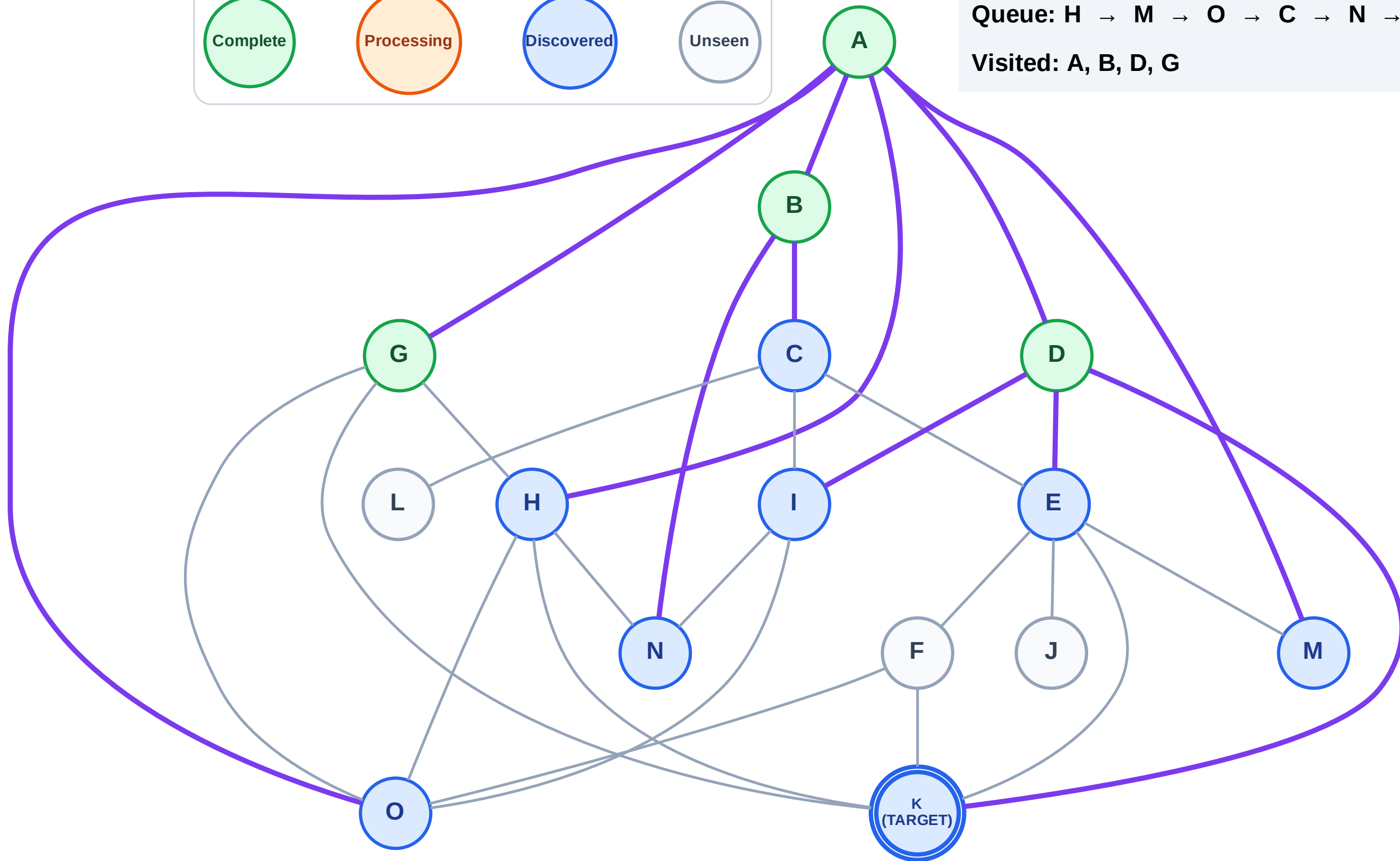
Step 9: No new neighbors from G

Legend



Queue: H → M → O → C → N → E → I → K

Visited: A, B, D, G



BFS Traversal

Step 10: Process node H

Legend

Complete

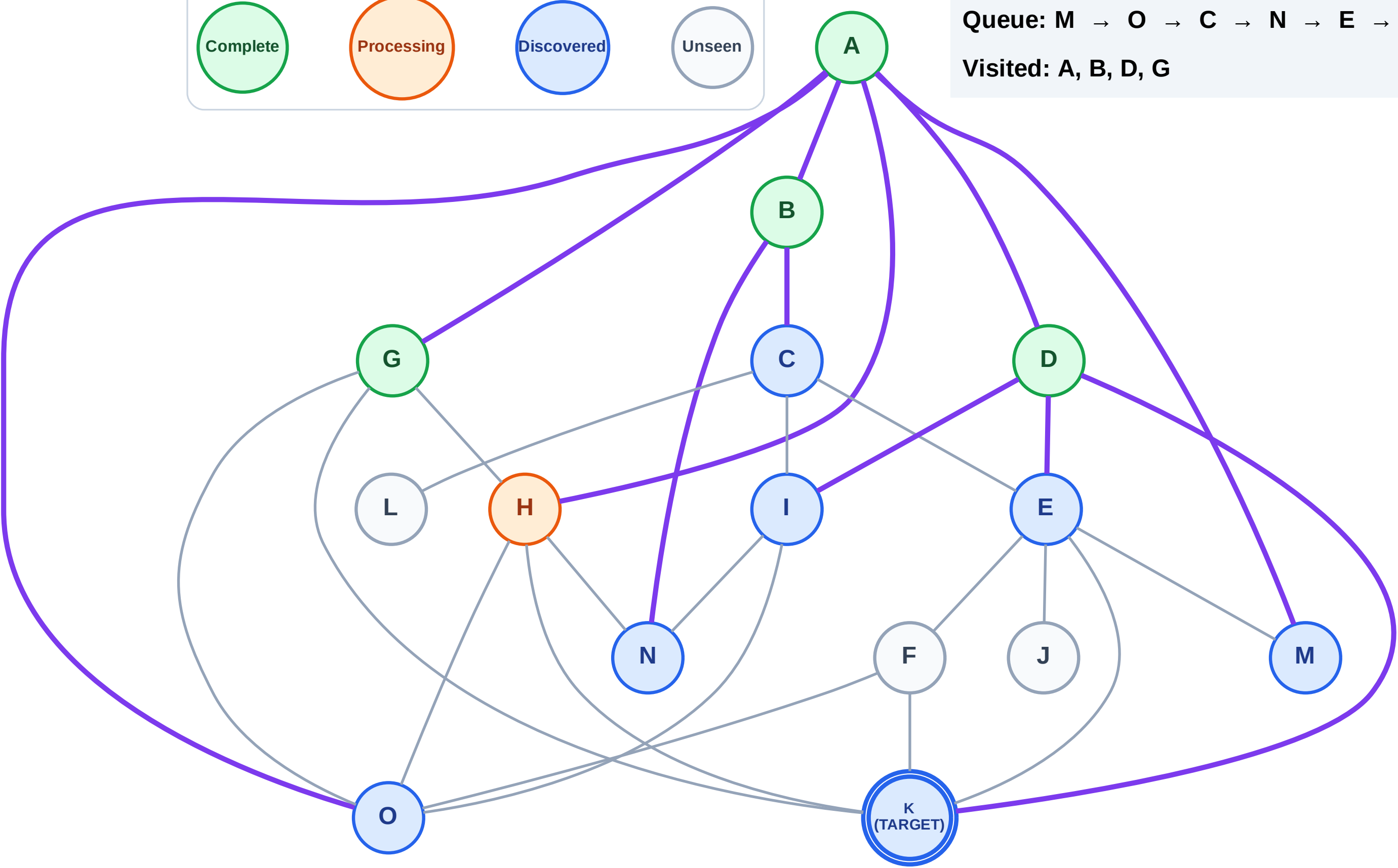
Processing

Discovered

Unseen

Queue: M → O → C → N → E → I → K

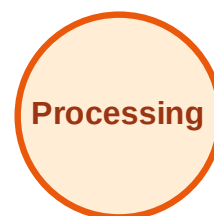
Visited: A, B, D, G



BFS Traversal

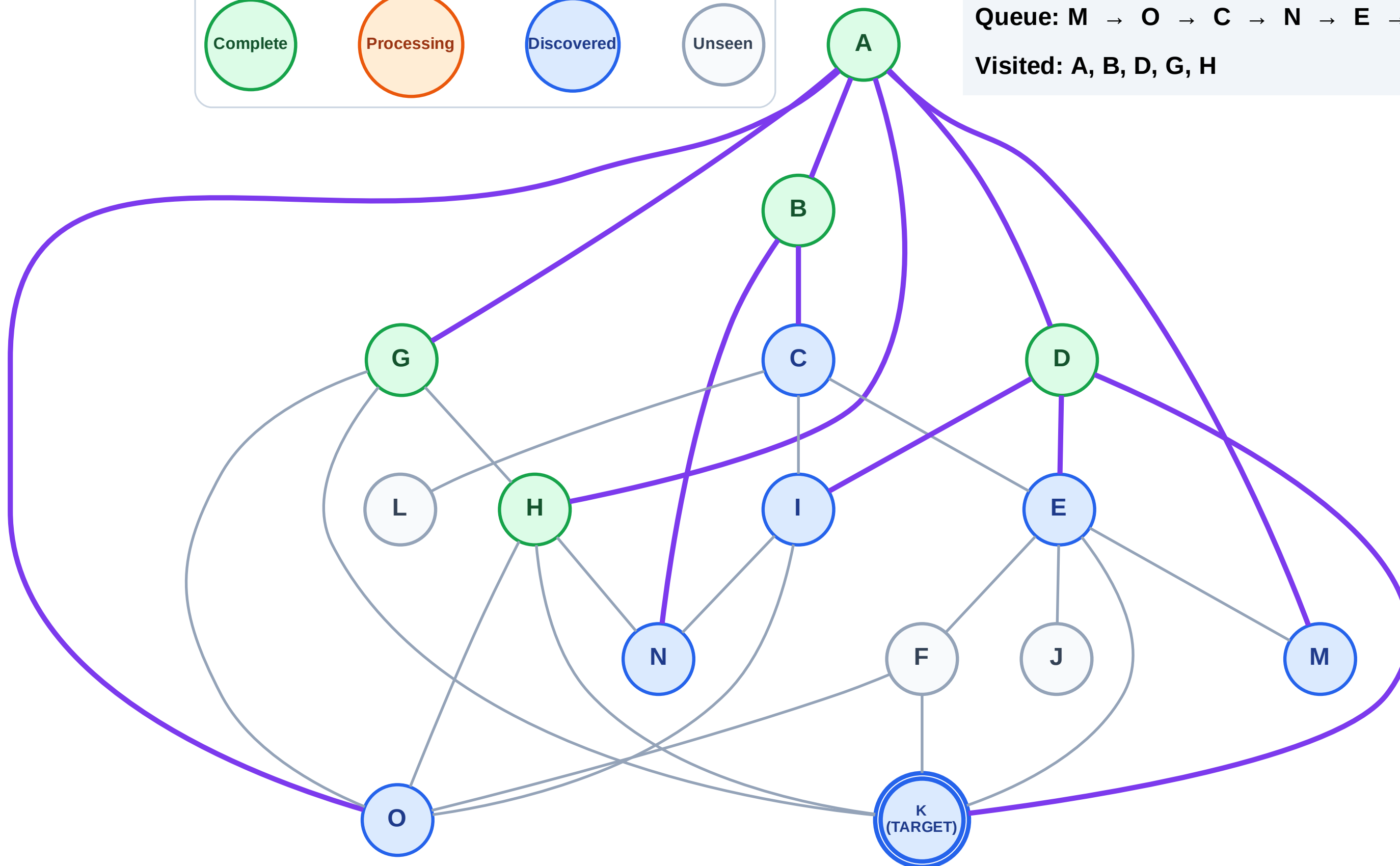
Step 11: No new neighbors from H

Legend



Queue: M → O → C → N → E → I → K

Visited: A, B, D, G, H



BFS Traversal

Step 12: Process node M

Legend

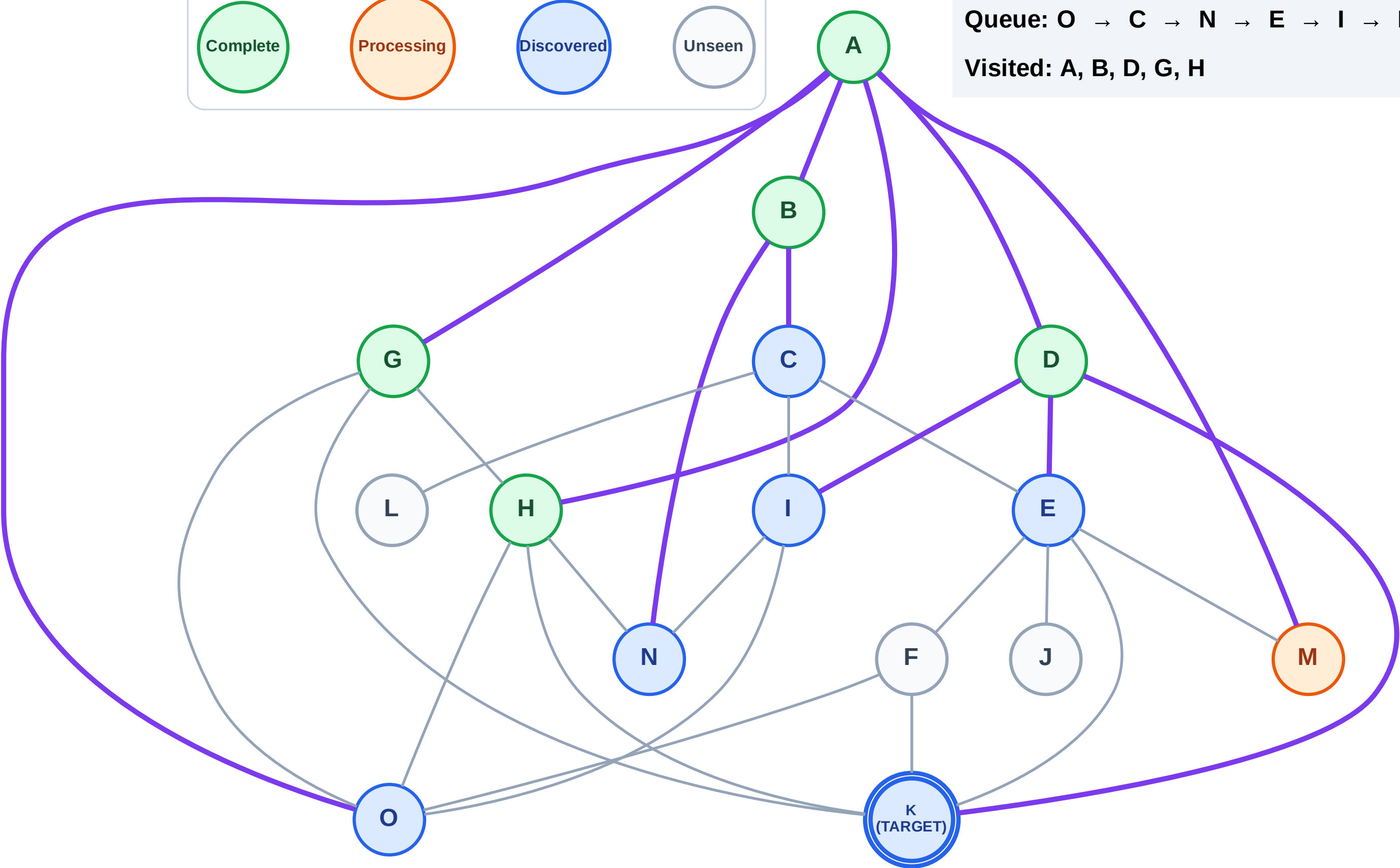
Complete

Processing

Discovered

Unseen

Queue: O → C → N → E → I → K
Visited: A, B, D, G, H



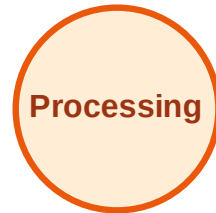
BFS Traversal

Step 13: No new neighbors from M

Legend



Complete



Processing



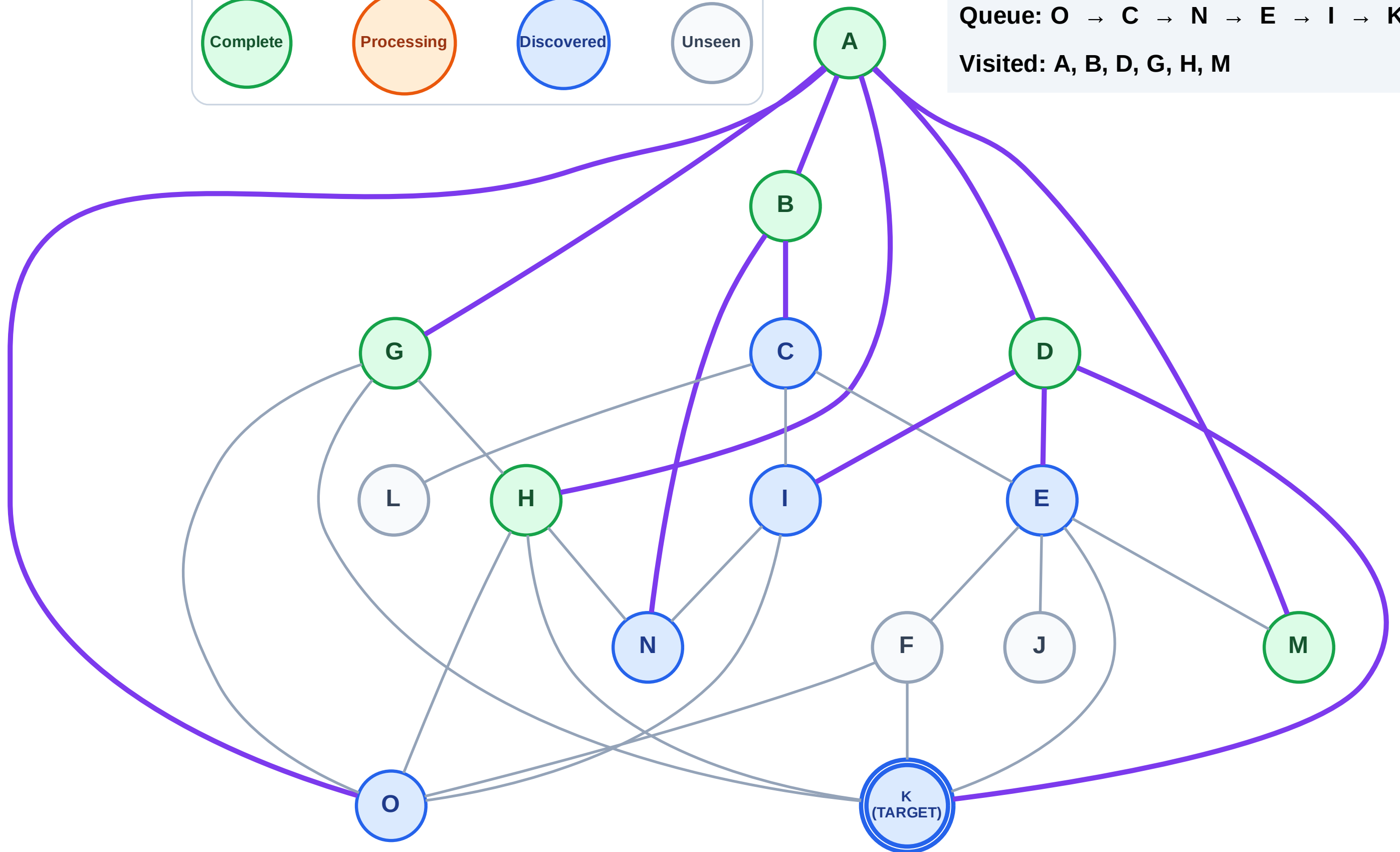
Discovered



Unseen

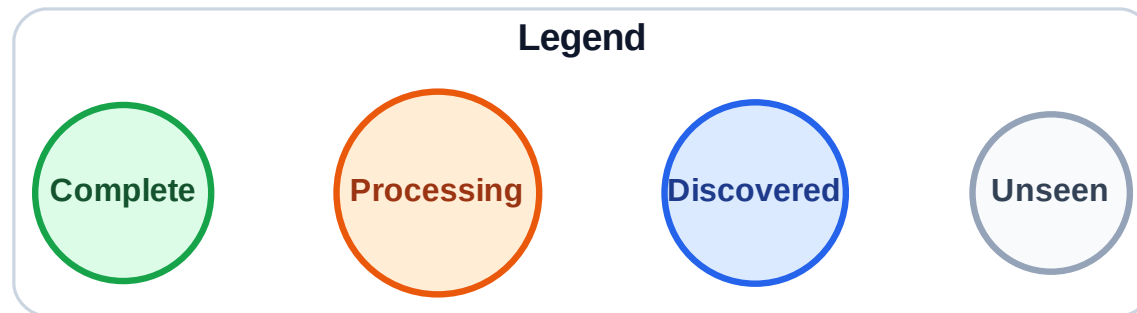
Queue: O → C → N → E → I → K

Visited: A, B, D, G, H, M

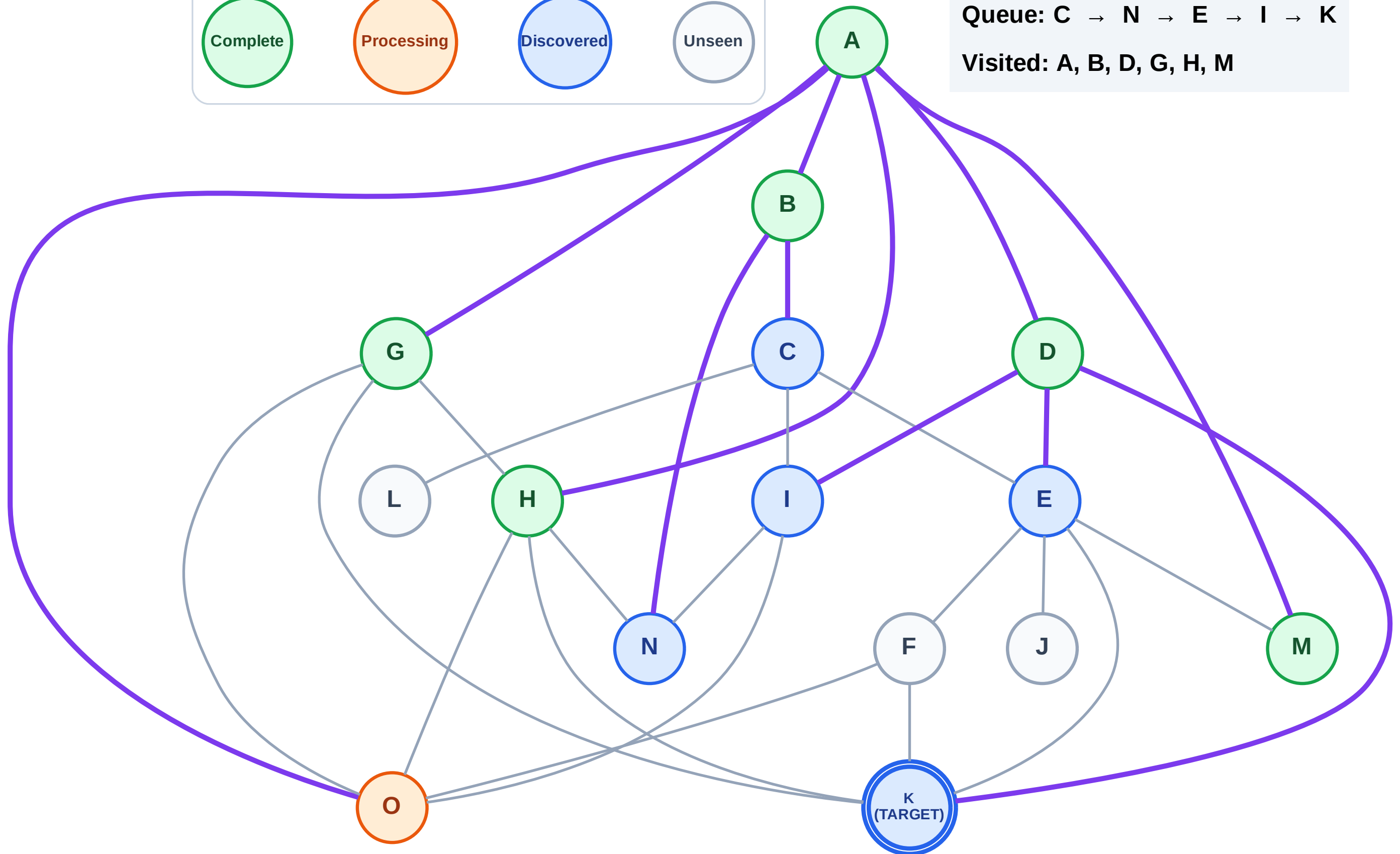


BFS Traversal

Step 14: Process node O

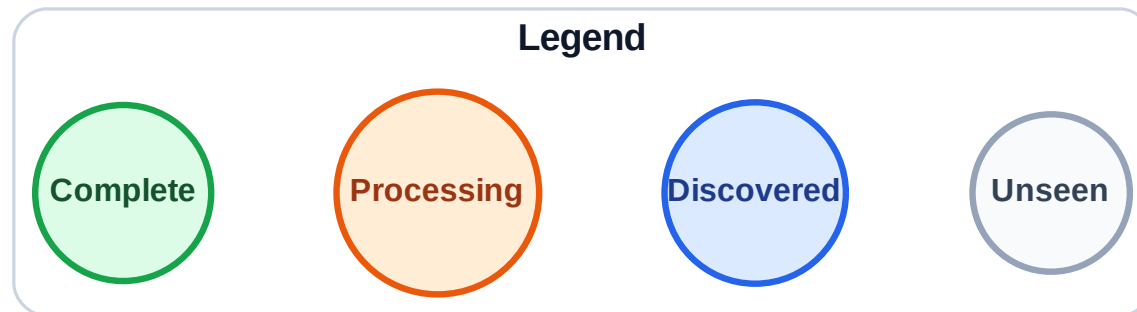


Queue: C → N → E → I → K
Visited: A, B, D, G, H, M

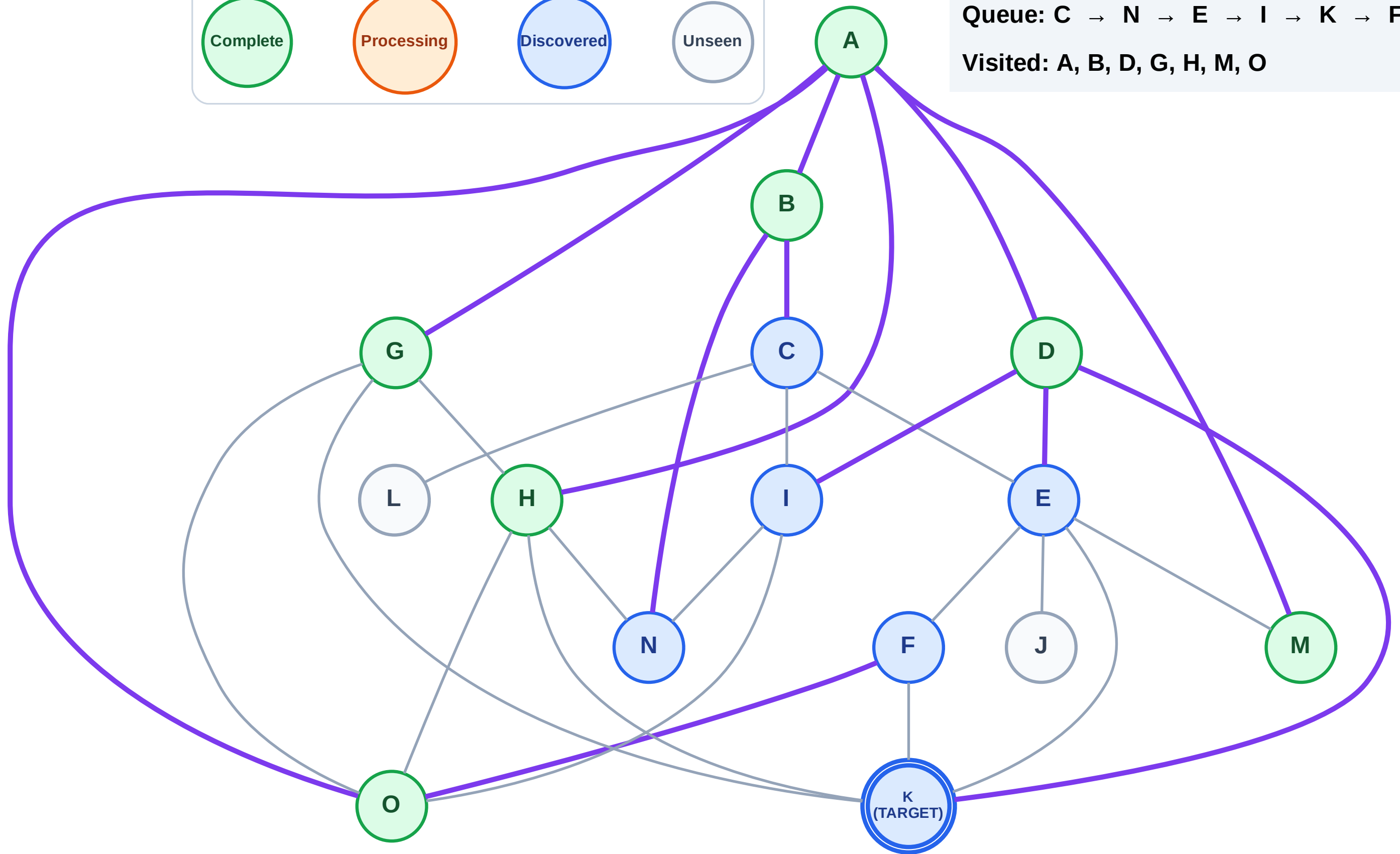


BFS Traversal

Step 15: Discover F from O

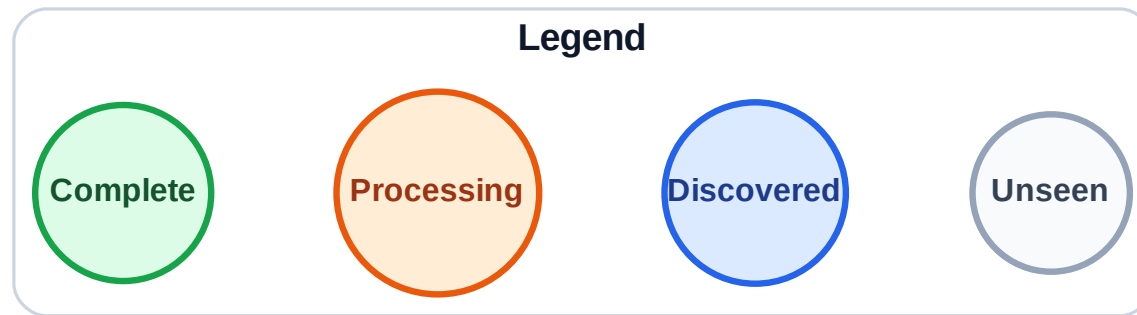


Queue: C → N → E → I → K → F
Visited: A, B, D, G, H, M, O



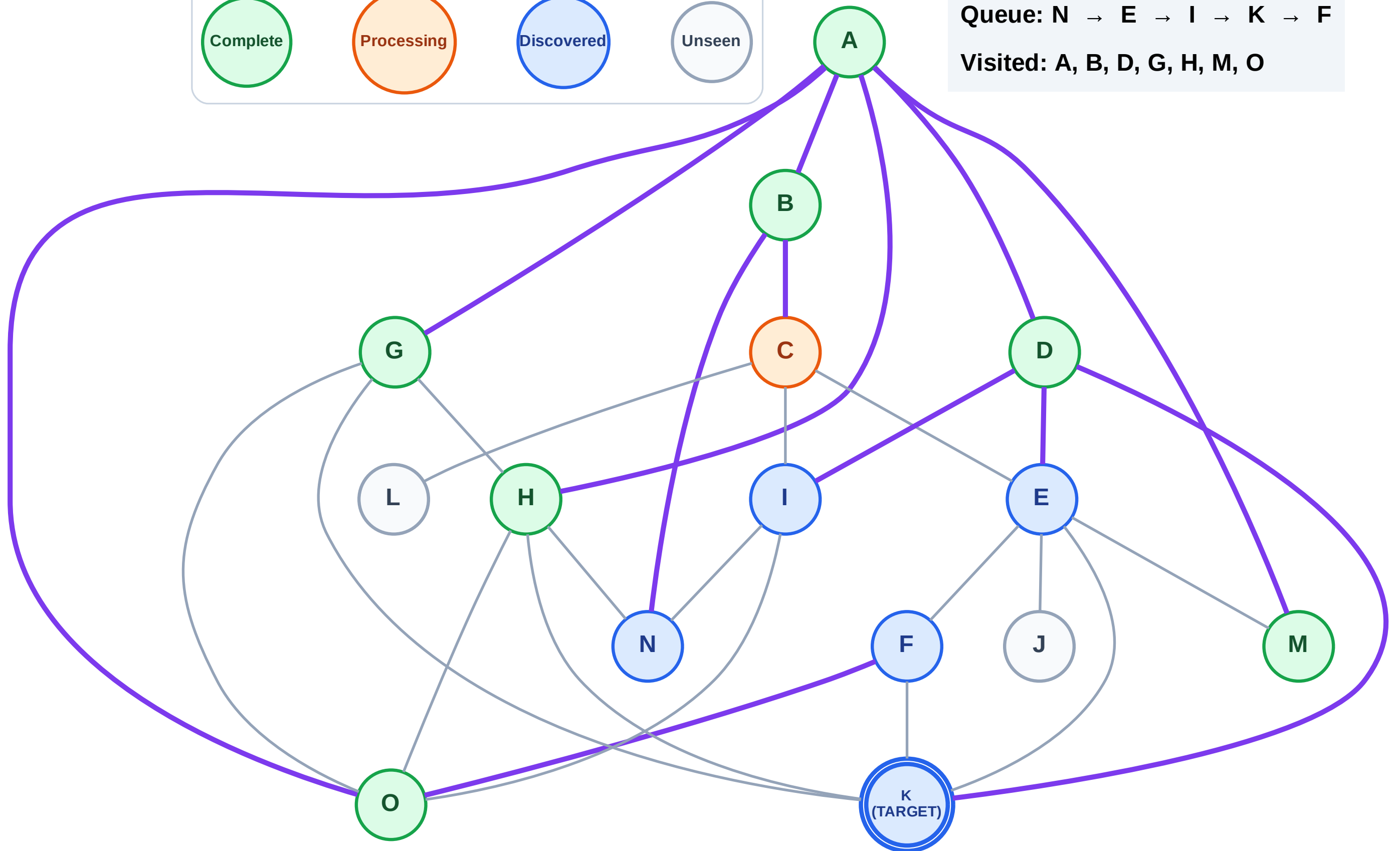
Step 16: Process node C

Step 16: Process node C



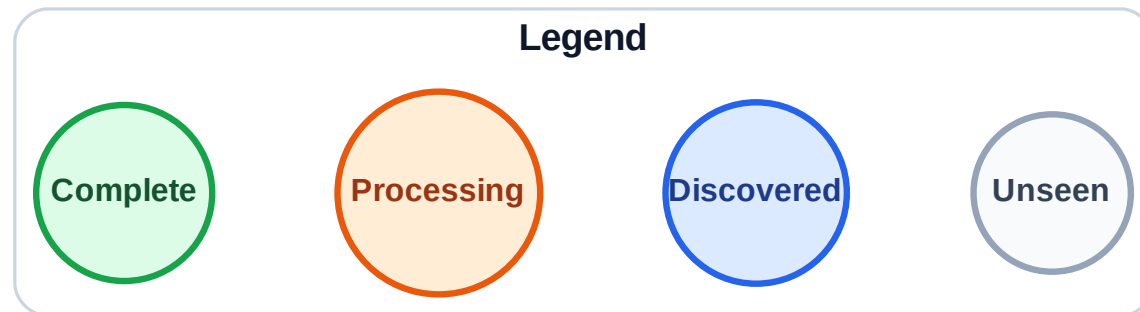
Queue: N → E → I → K → F

Visited: A, B, D, G, H, M, O



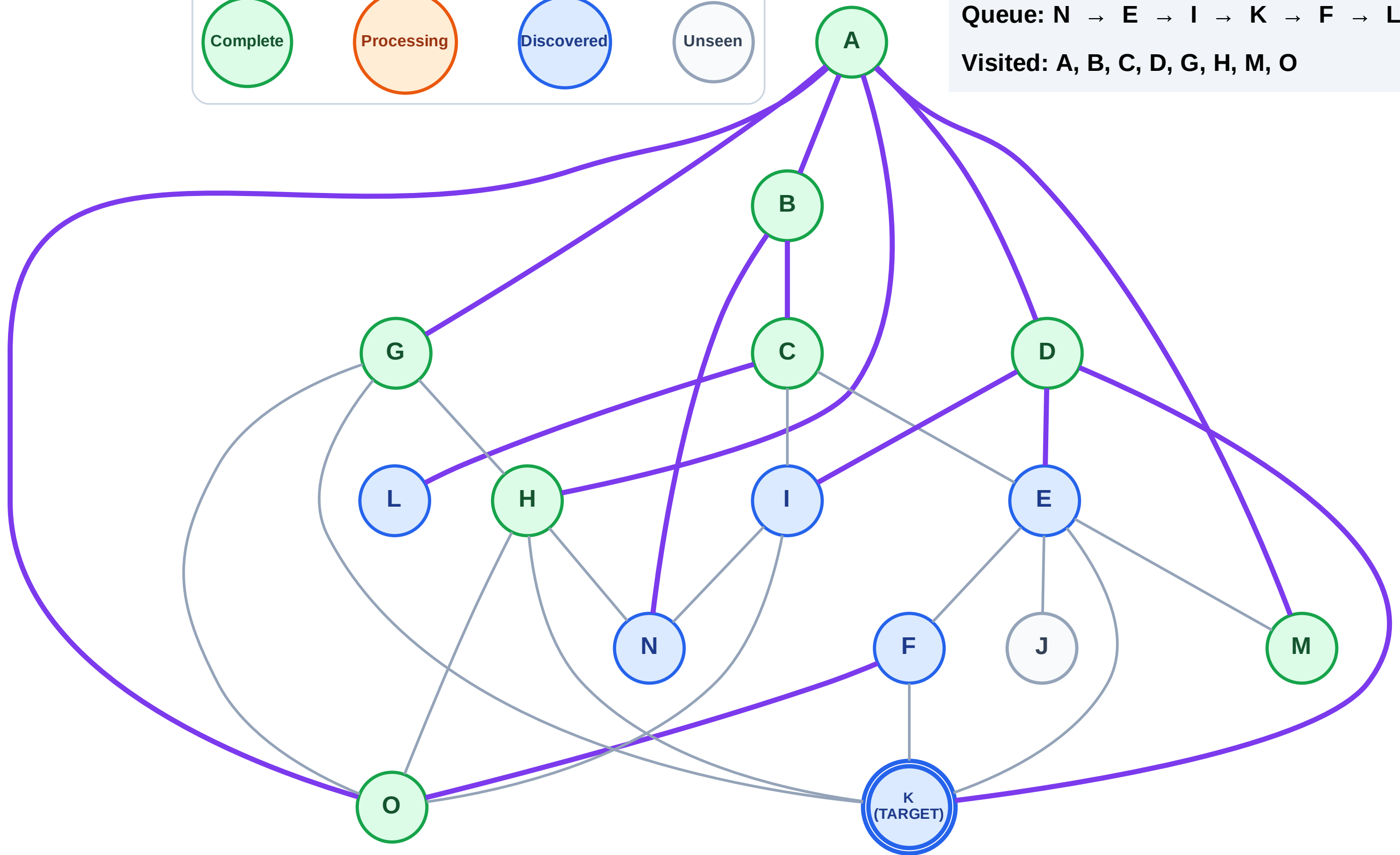
BFS Traversal

Step 17: Discover L from C



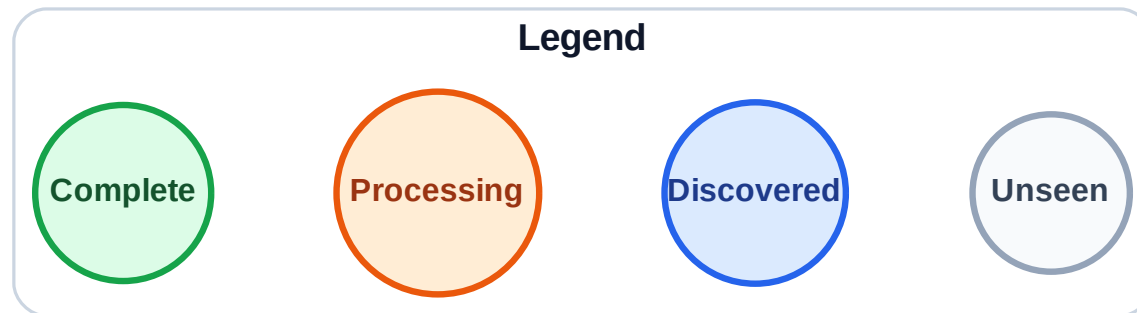
Queue: N → E → I → K → F → L

Visited: A, B, C, D, G, H, M, O



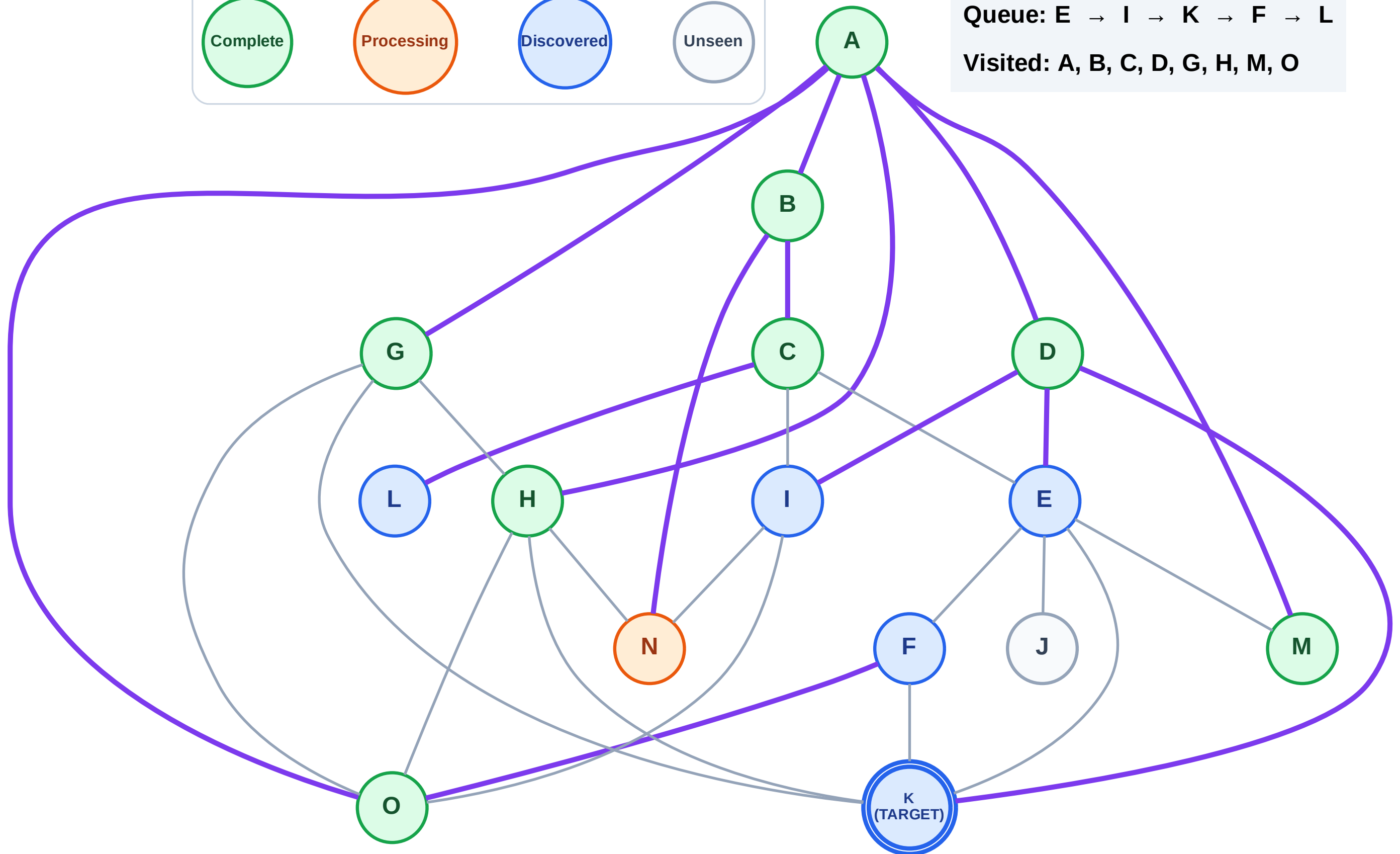
BFS Traversal

Step 18: Process node N



Queue: E → I → K → F → L

Visited: A, B, C, D, G, H, M, O



BFS Traversal

Step 19: No new neighbors from N

Legend

Complete

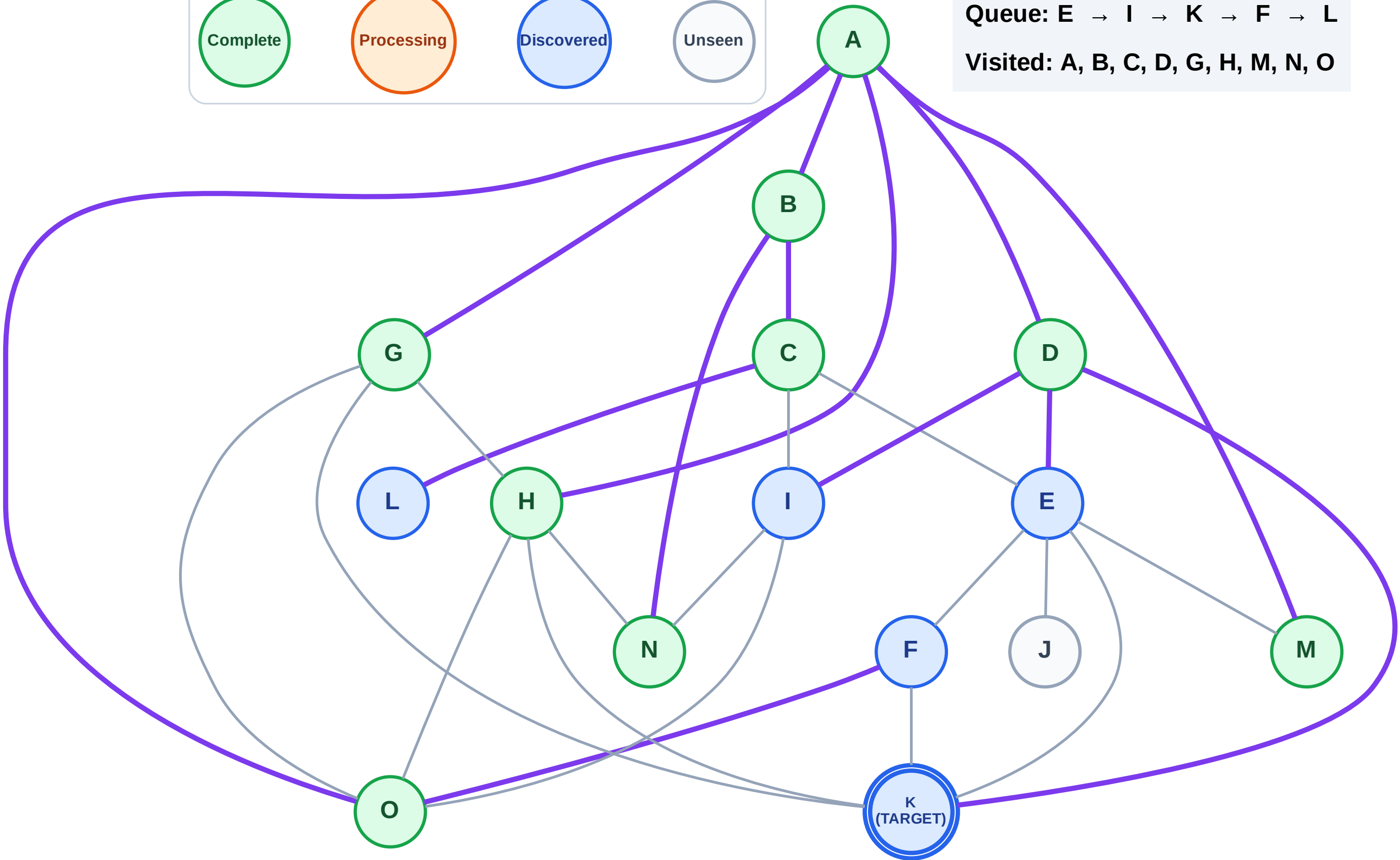
Processing

Discovered

Unseen

Queue: E → I → K → F → L

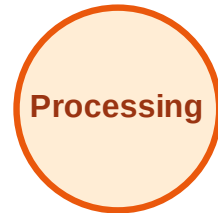
Visited: A, B, C, D, G, H, M, N, O



BFS Traversal

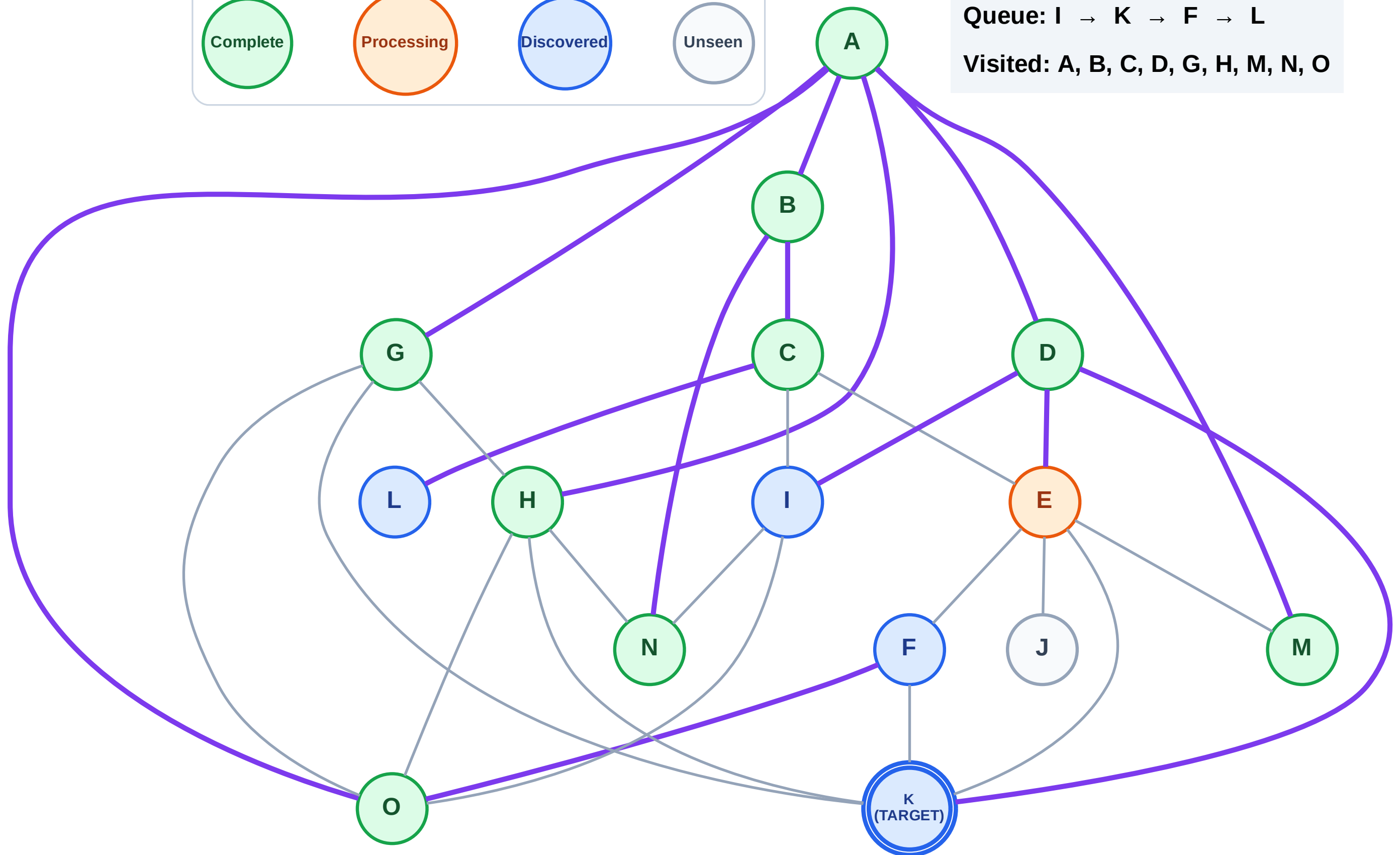
Step 20: Process node E

Legend



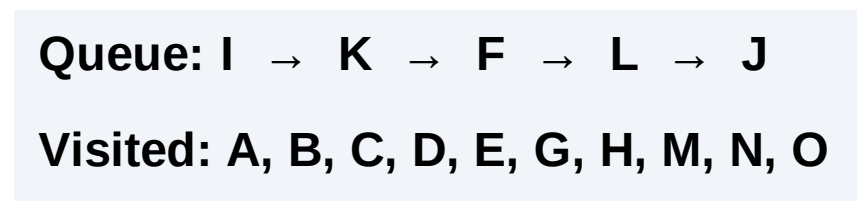
Queue: I → K → F → L

Visited: A, B, C, D, G, H, M, N, O



Step 21: Discover J from E

Step 21: Discover J from E



BFS Traversal

Step 22: Process node I

Legend

Complete

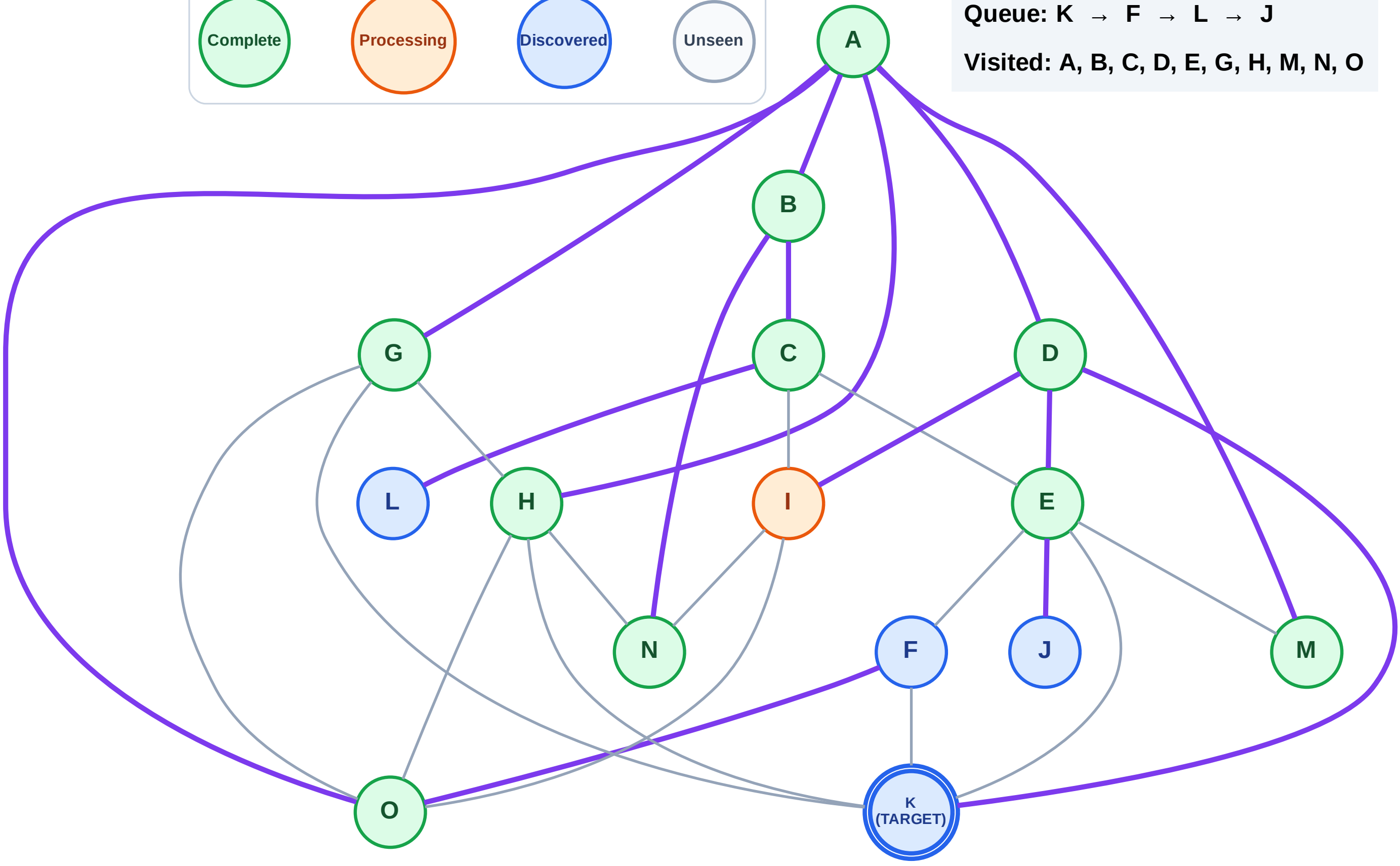
Processing

Discovered

Unseen

Queue: K → F → L → J

Visited: A, B, C, D, E, G, H, M, N, O



BFS Traversal

Step 23: No new neighbors from I

Legend

Complete

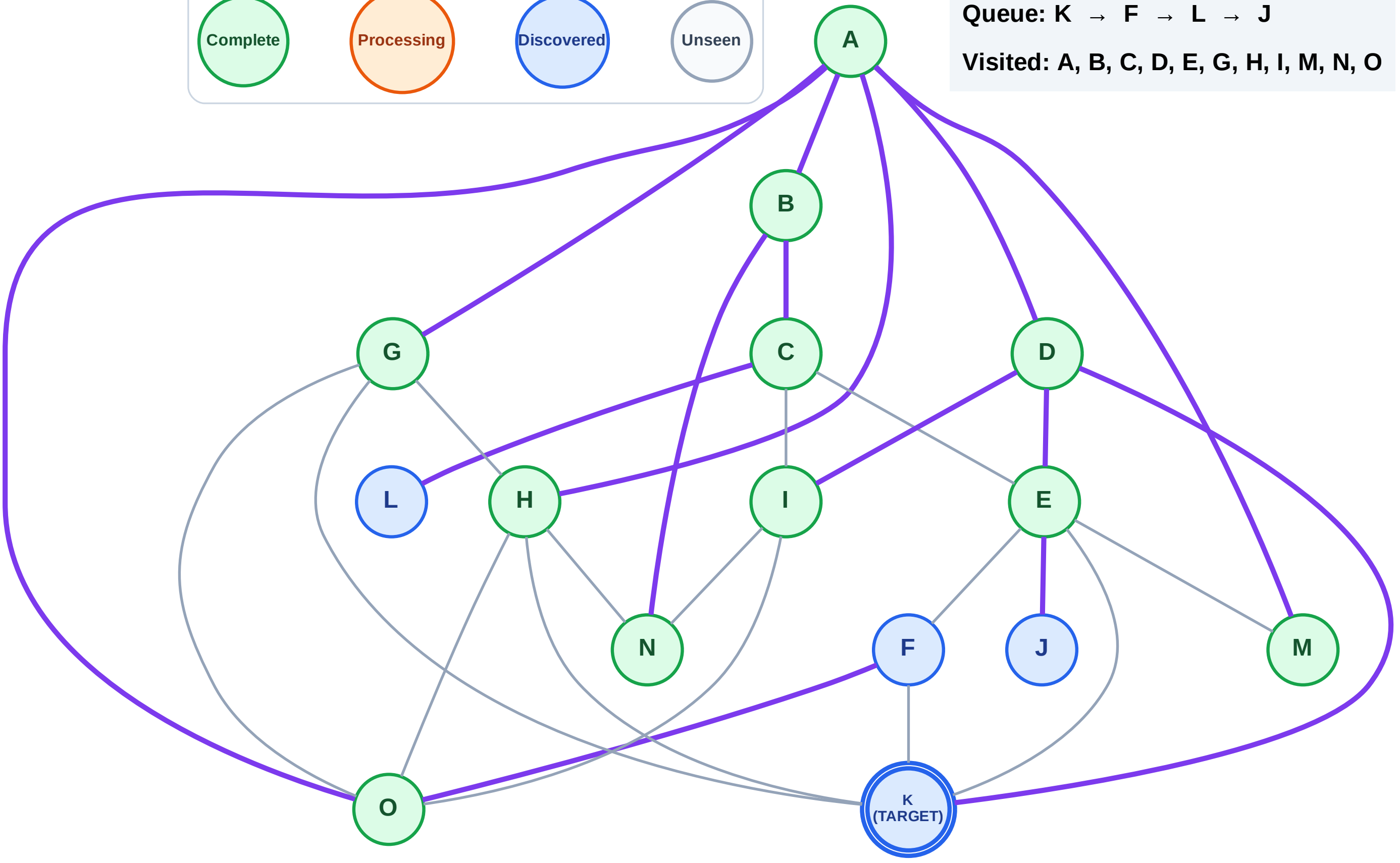
Processing

Discovered

Unseen

Queue: K → F → L → J

Visited: A, B, C, D, E, G, H, I, M, N, O



BFS Traversal

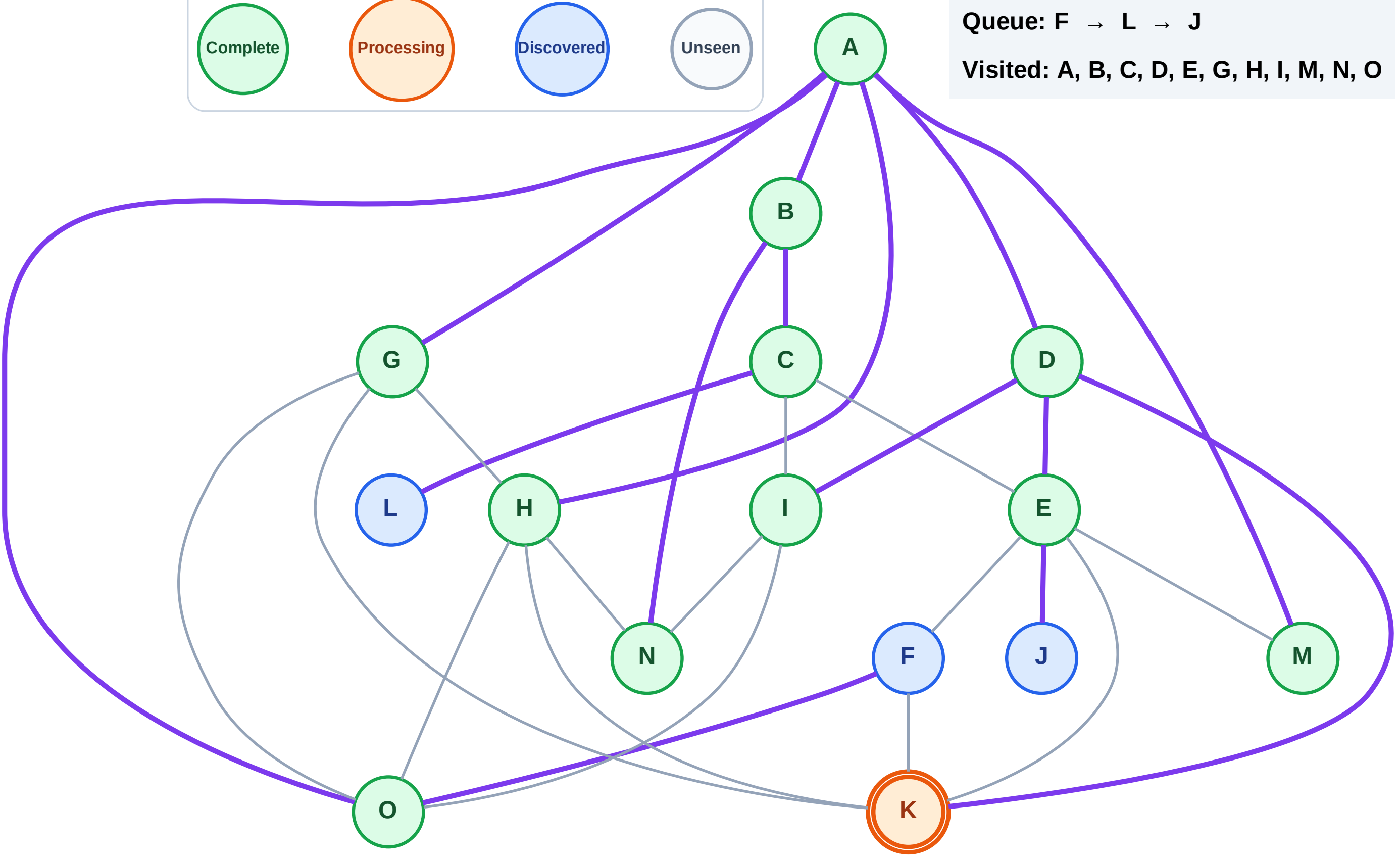
Step 24: Process node K

Legend



Queue: F → L → J

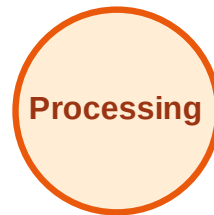
Visited: A, B, C, D, E, G, H, I, M, N, O



BFS Traversal

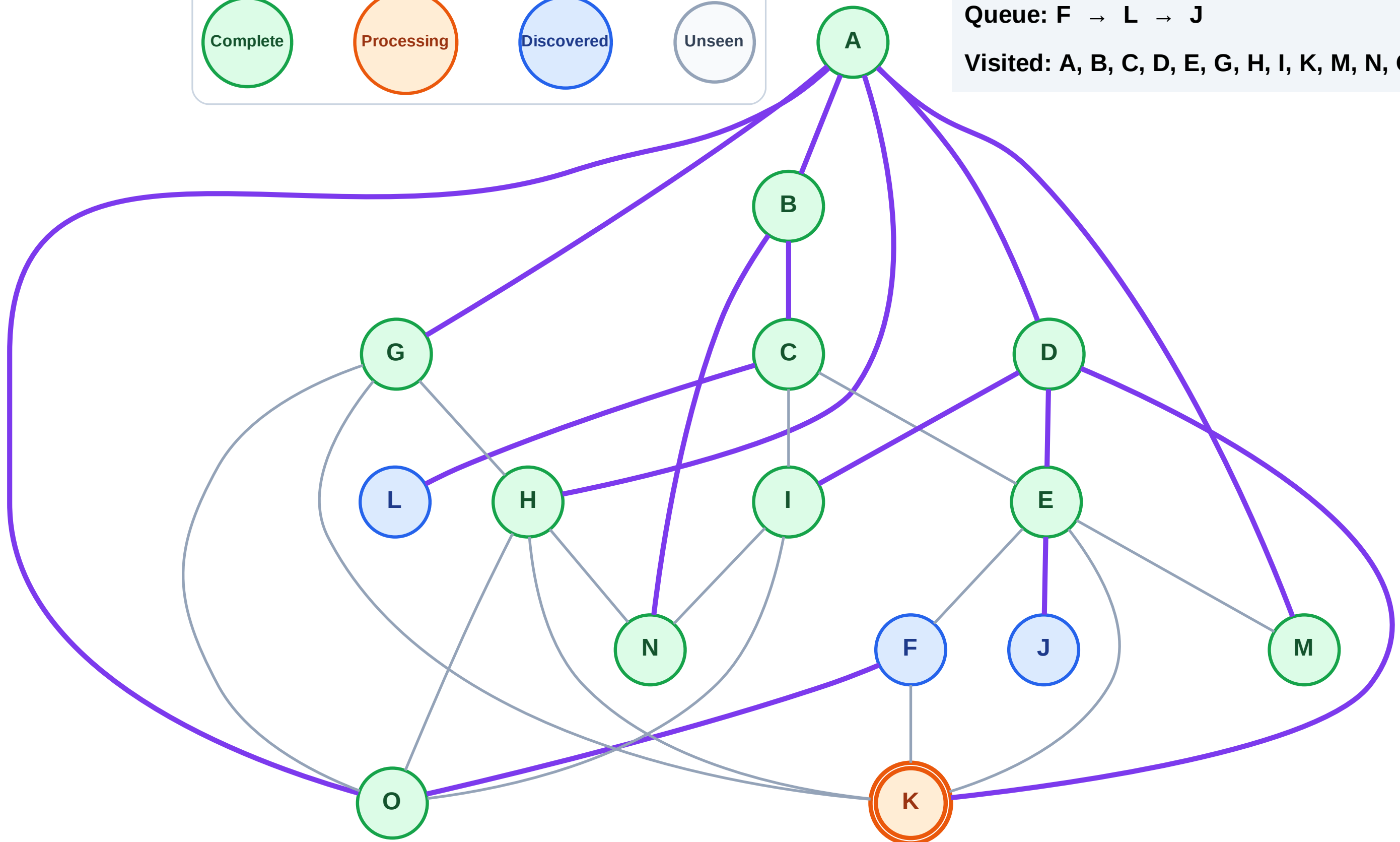
Step 25: TARGET NODE K REACHED! Path (2 hops): A → D → K

Legend



Queue: F → L → J

Visited: A, B, C, D, E, G, H, I, K, M, N, O



DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

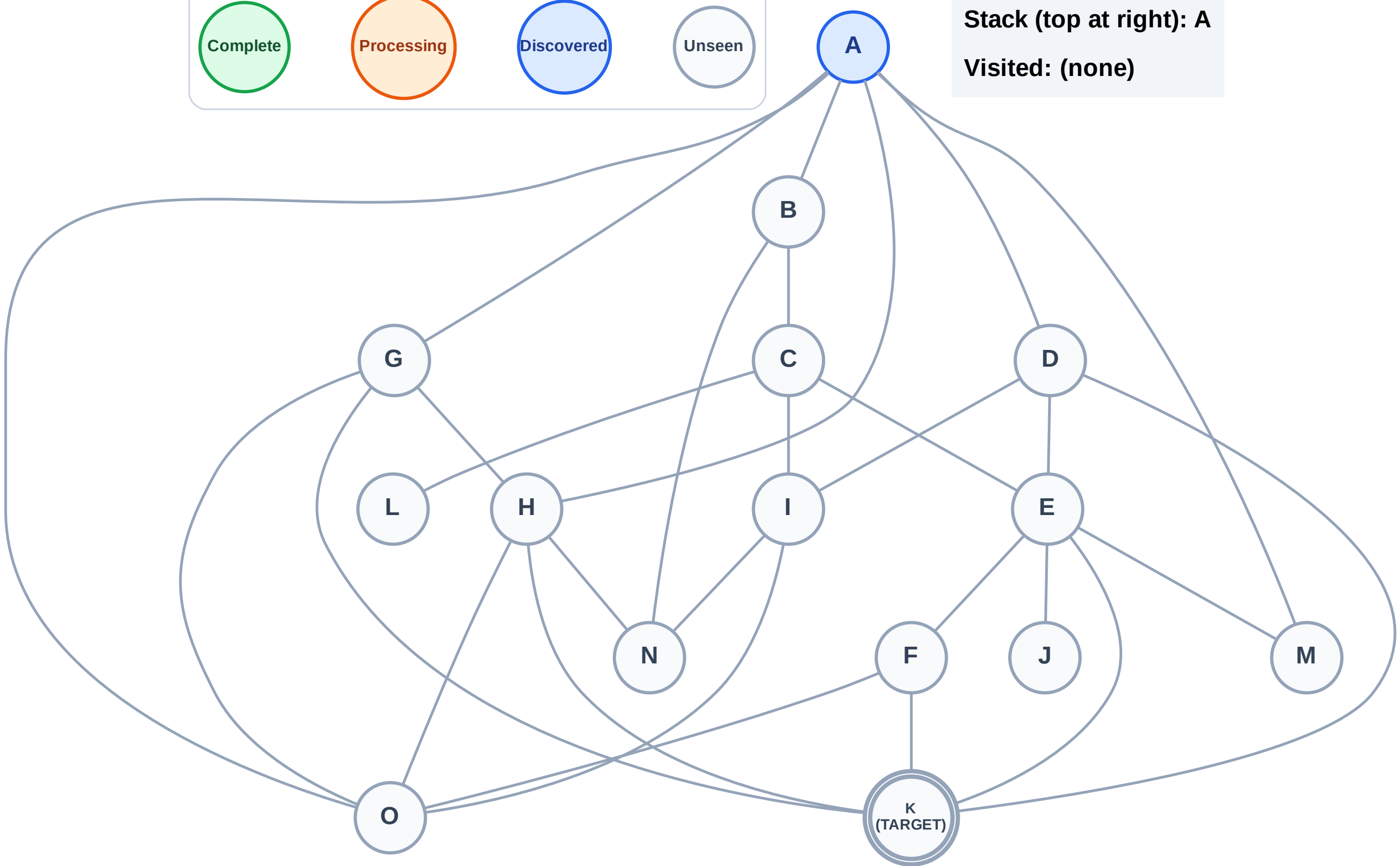
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

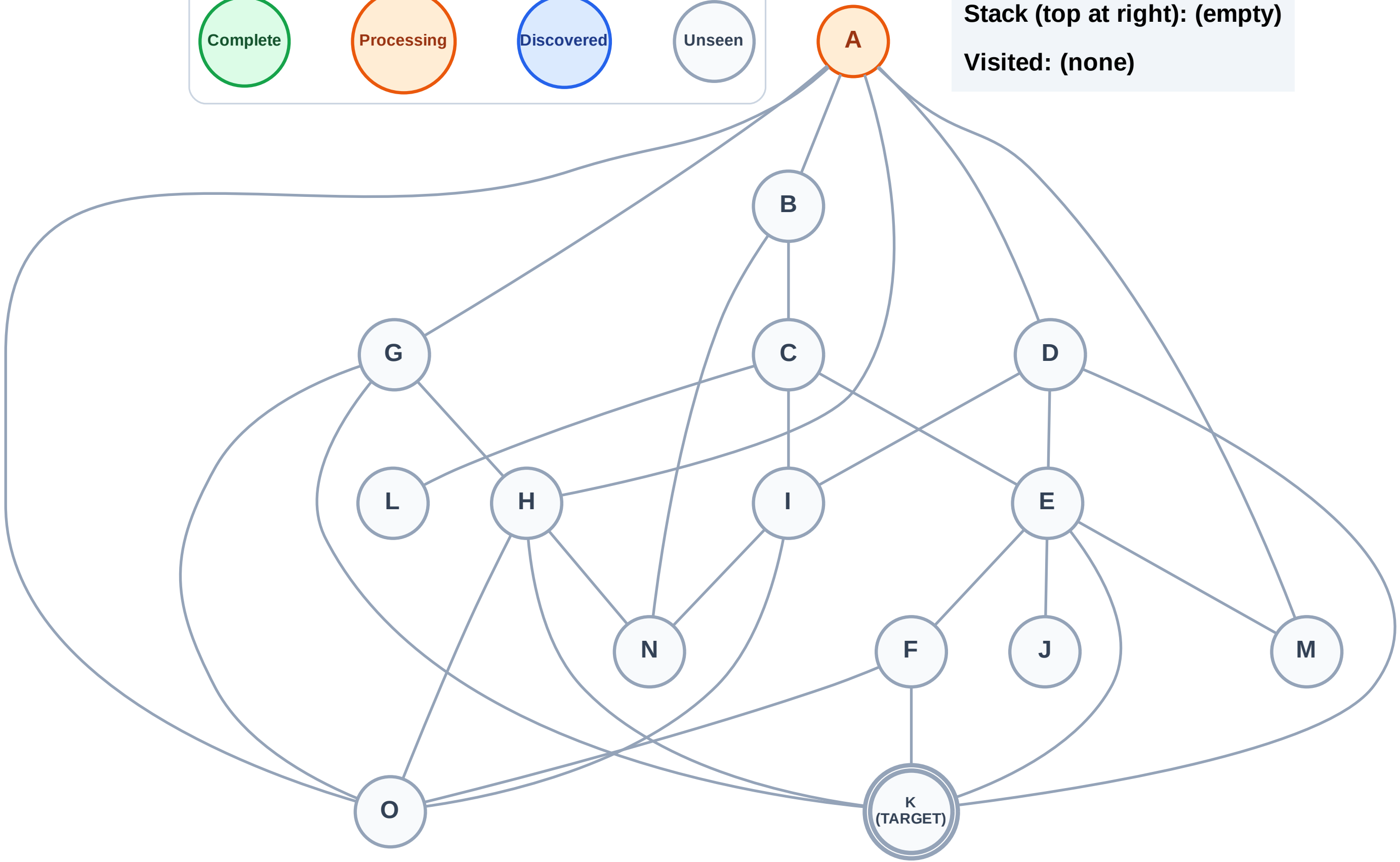
Complete

Processing

Discovered

Unseen

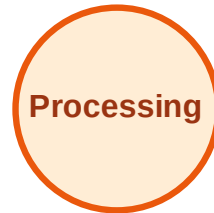
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

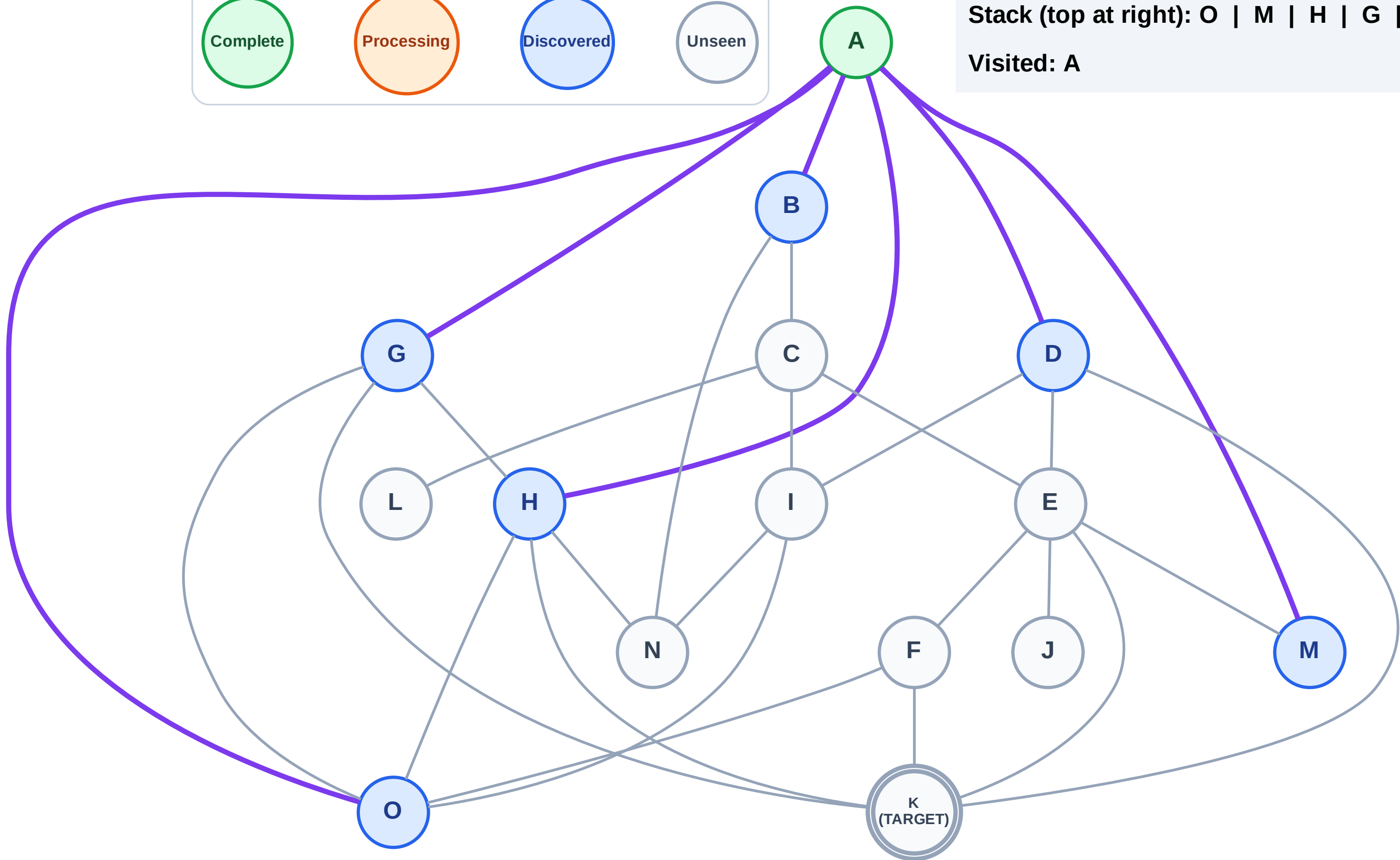
Step 3: Discover O, M, H, G, D, B from A

Legend

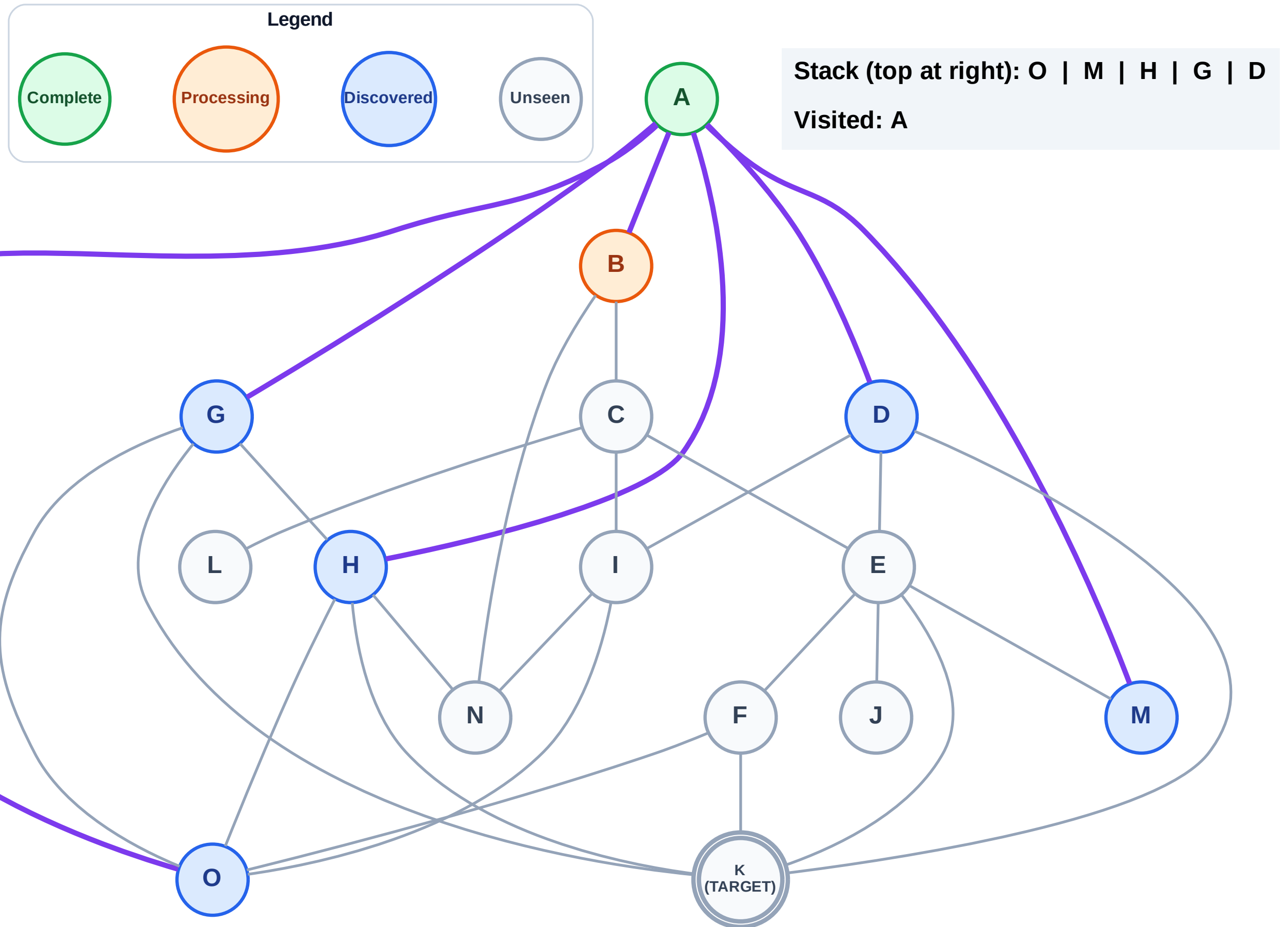


Stack (top at right): O | M | H | G | D | B

Visited: A

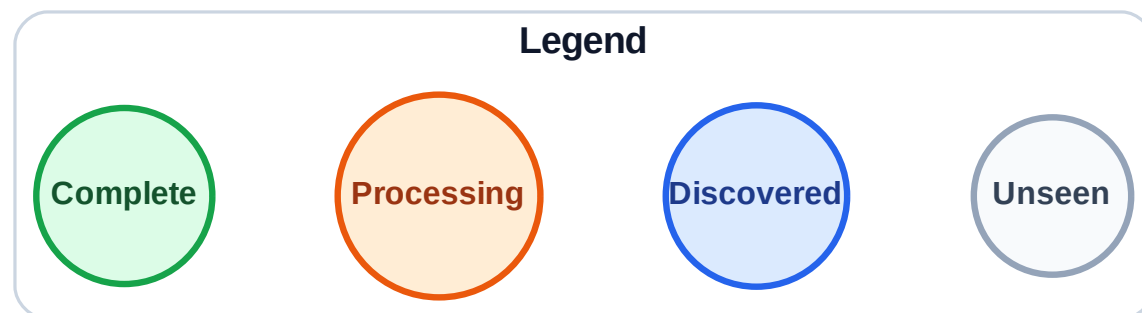


Step 4: Process node B



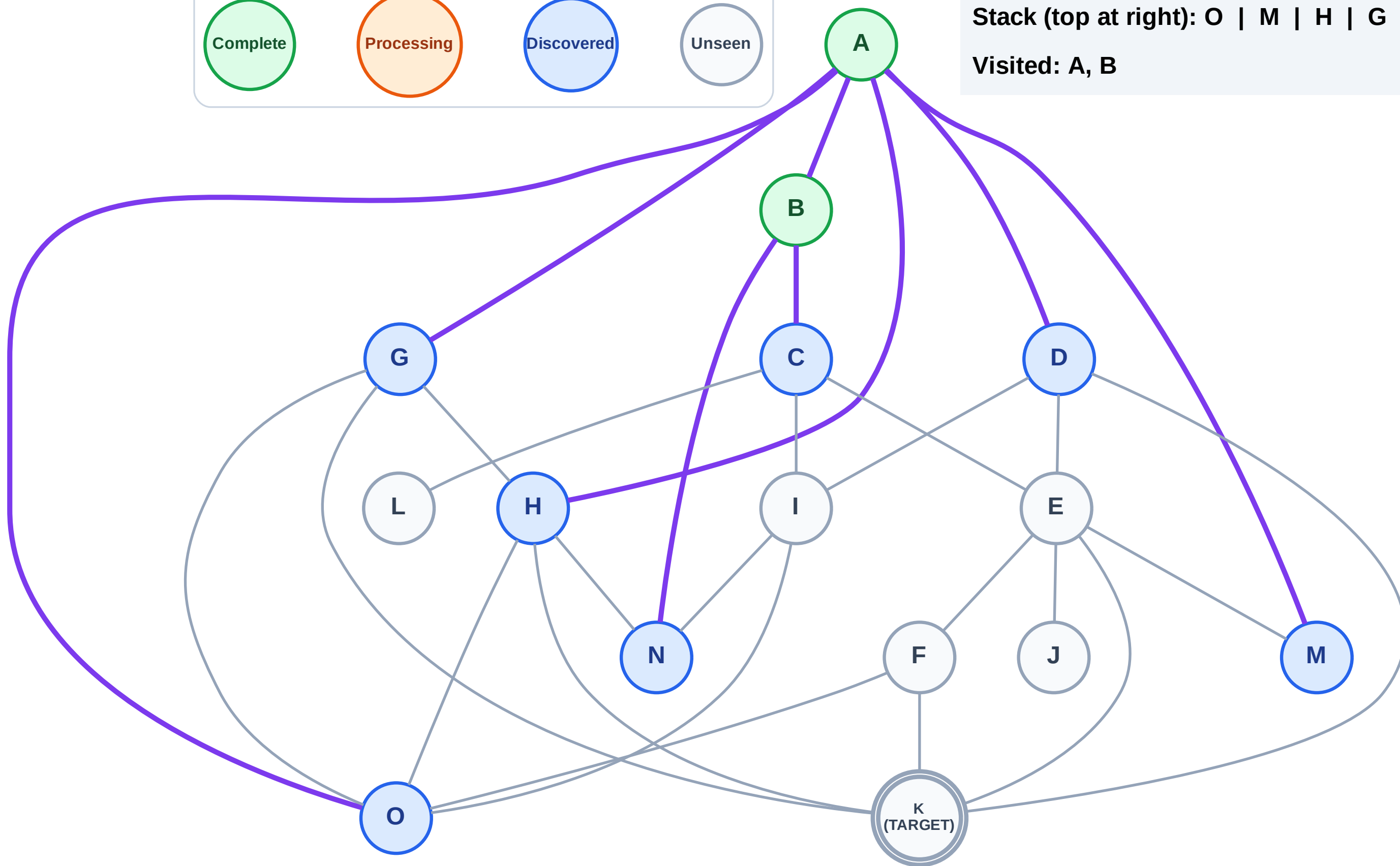
DFS Traversal

Step 5: Discover N, C from B



Stack (top at right): O | M | H | G | D | N | C

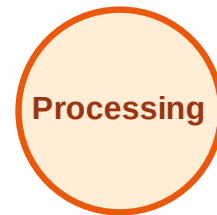
Visited: A, B



DFS Traversal

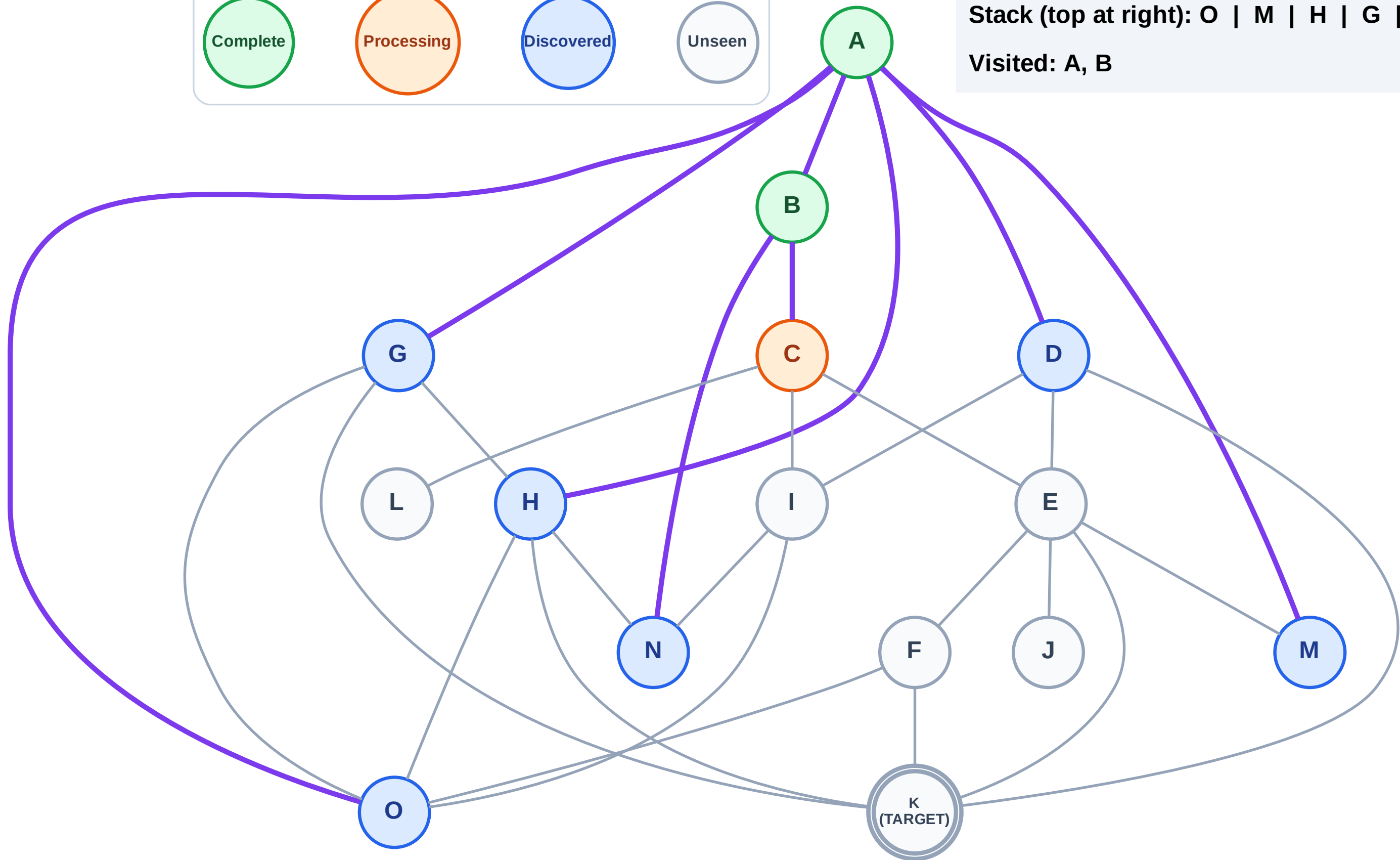
Step 6: Process node C

Legend



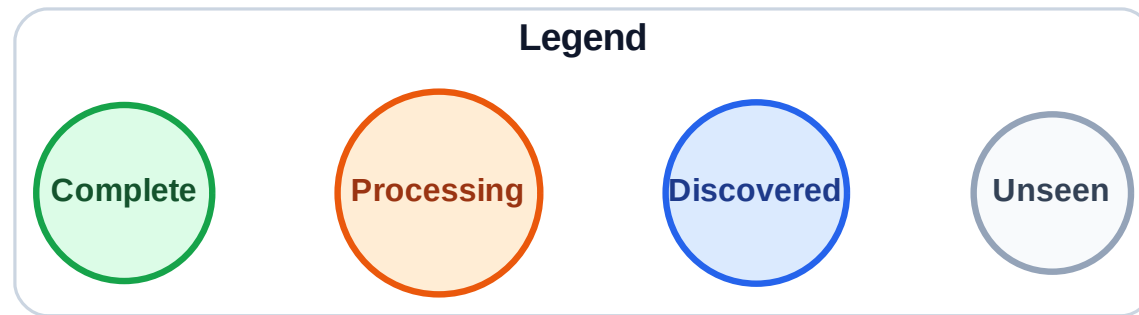
Stack (top at right): O | M | H | G | D | N

Visited: A, B



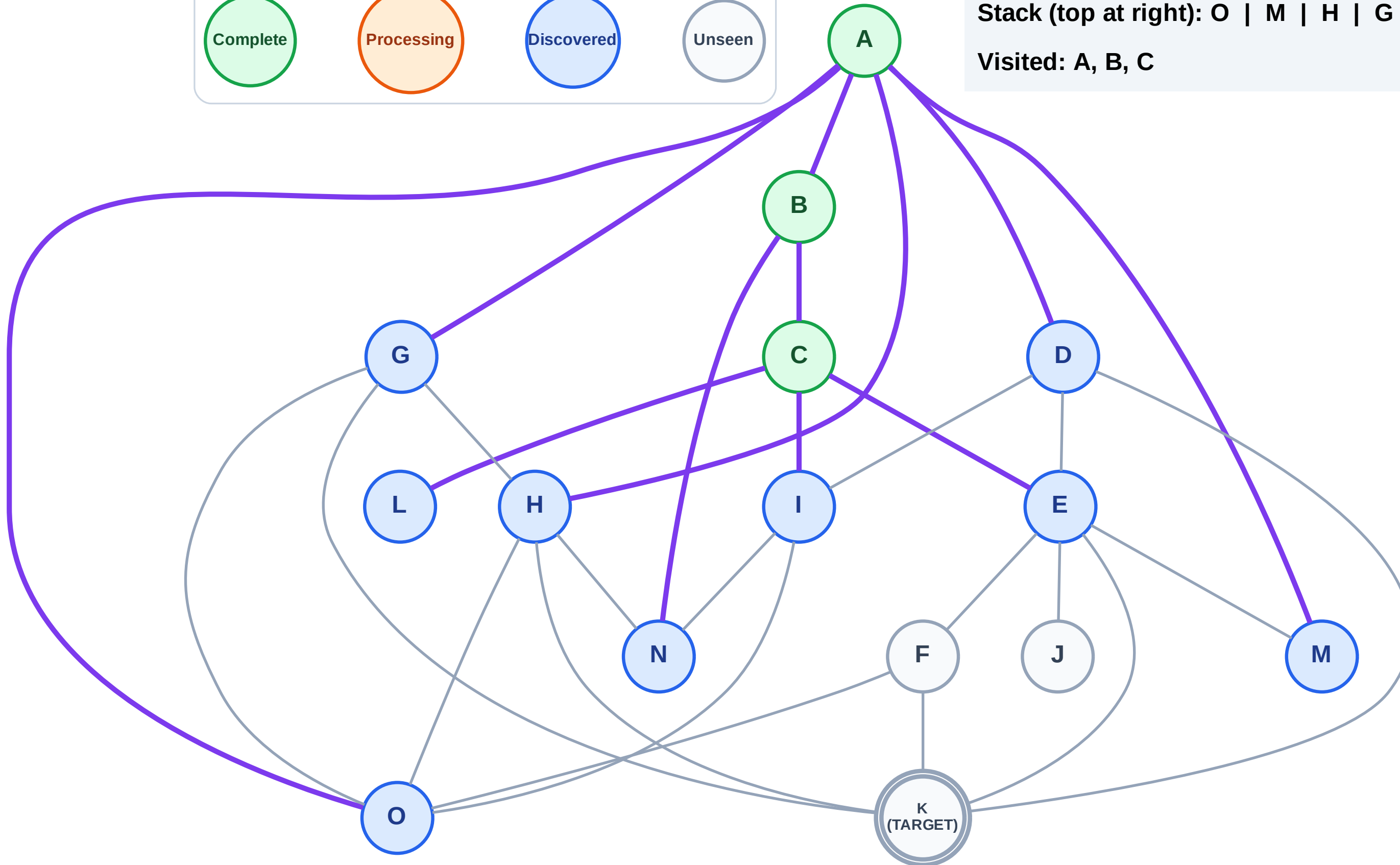
Step 7: Discover L, I, E from C

Step 7: Discover L, I, E from C



Stack (top at right): O | M | H | G | D | N | L | I | E

Visited: A, B, C



DFS Traversal

Step 8: Process node E

Legend

Complete

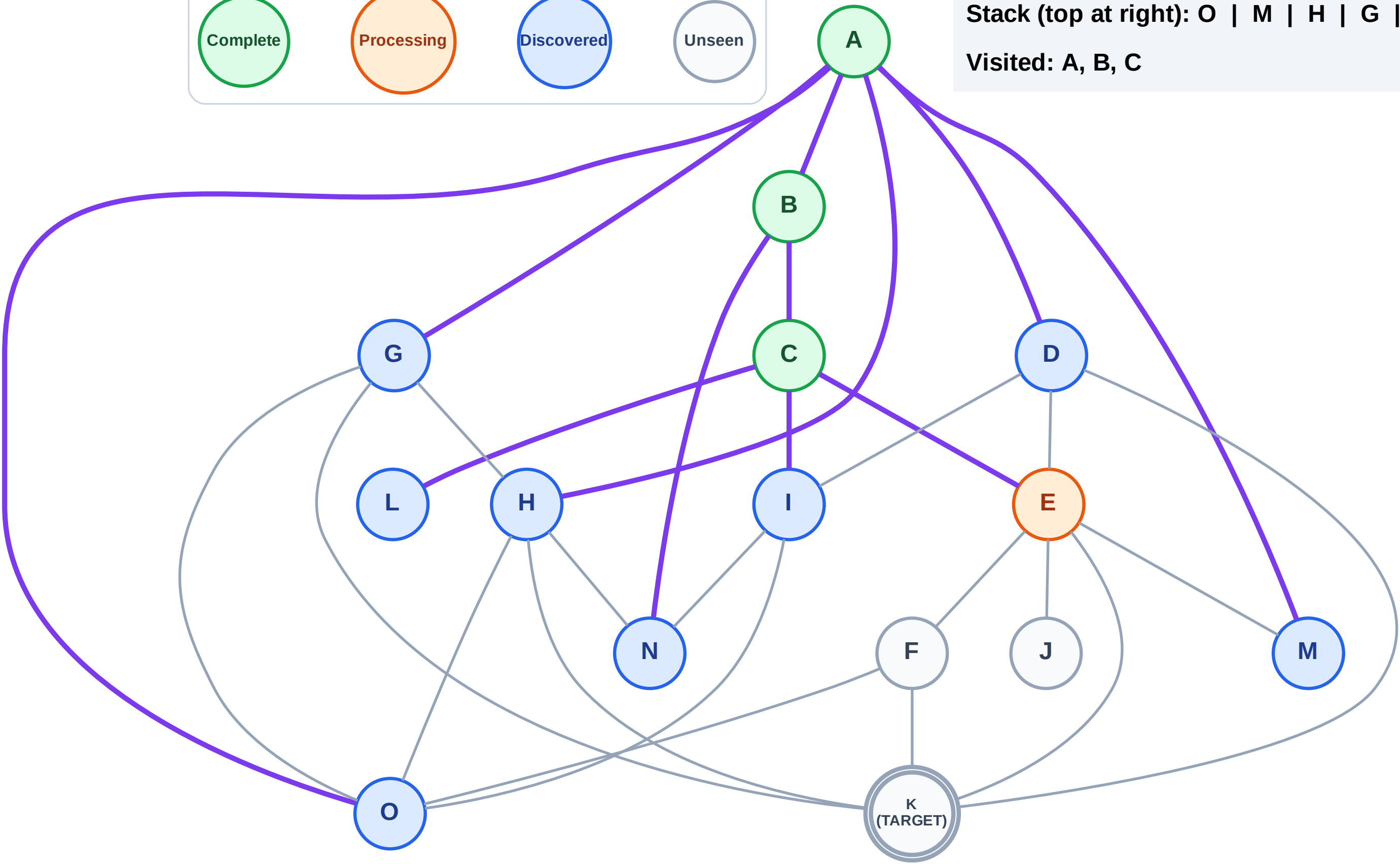
Processing

Discovered

Unseen

Stack (top at right): O | M | H | G | D | N | L | I

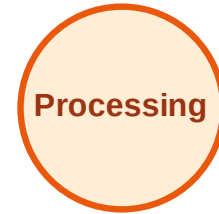
Visited: A, B, C



DFS Traversal

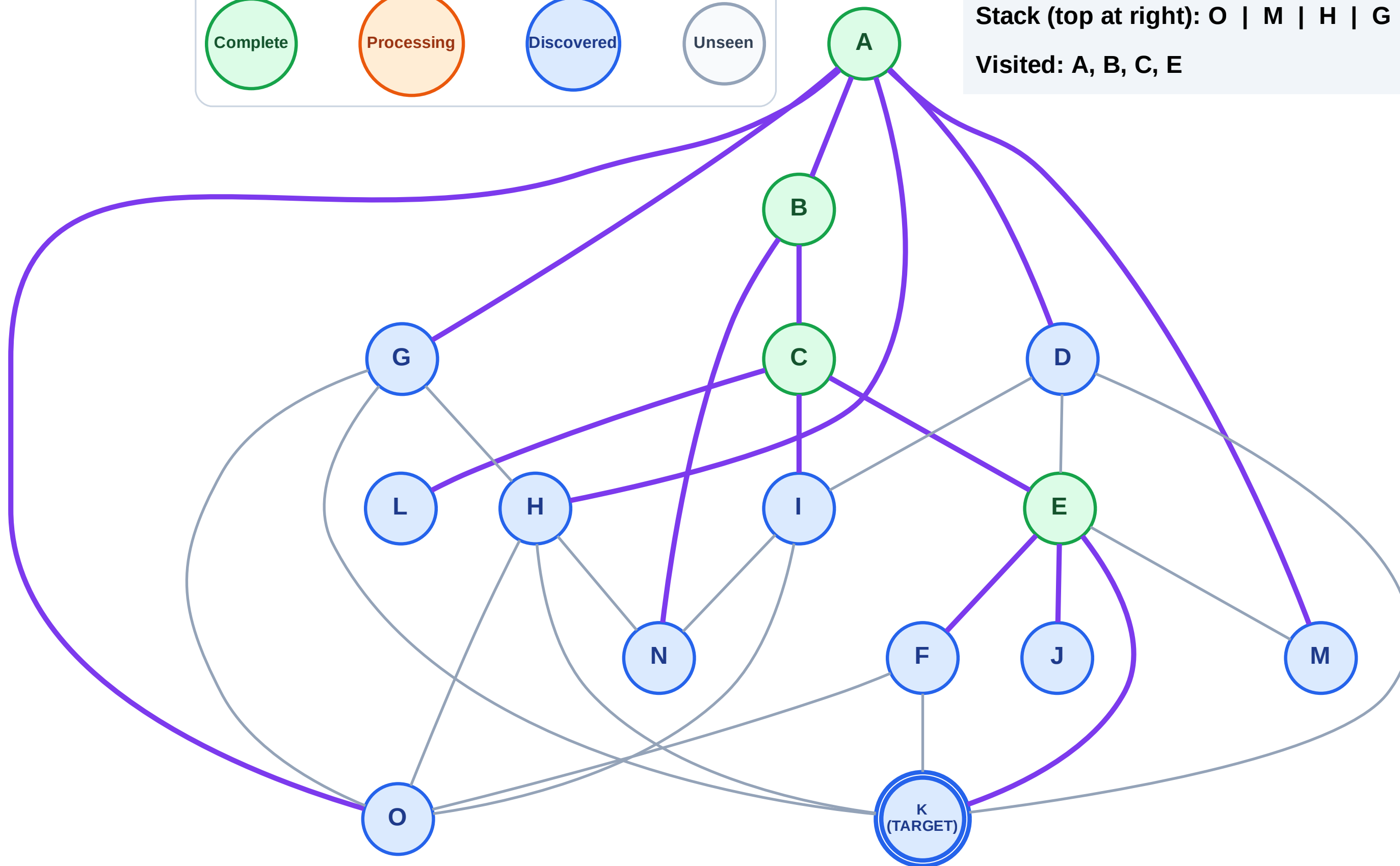
Step 9: Discover K, J, F from E

Legend



Stack (top at right): O | M | H | G | D | N | L | I | K | J | F

Visited: A, B, C, E



DFS Traversal

Step 10: Process node F

Legend

Complete

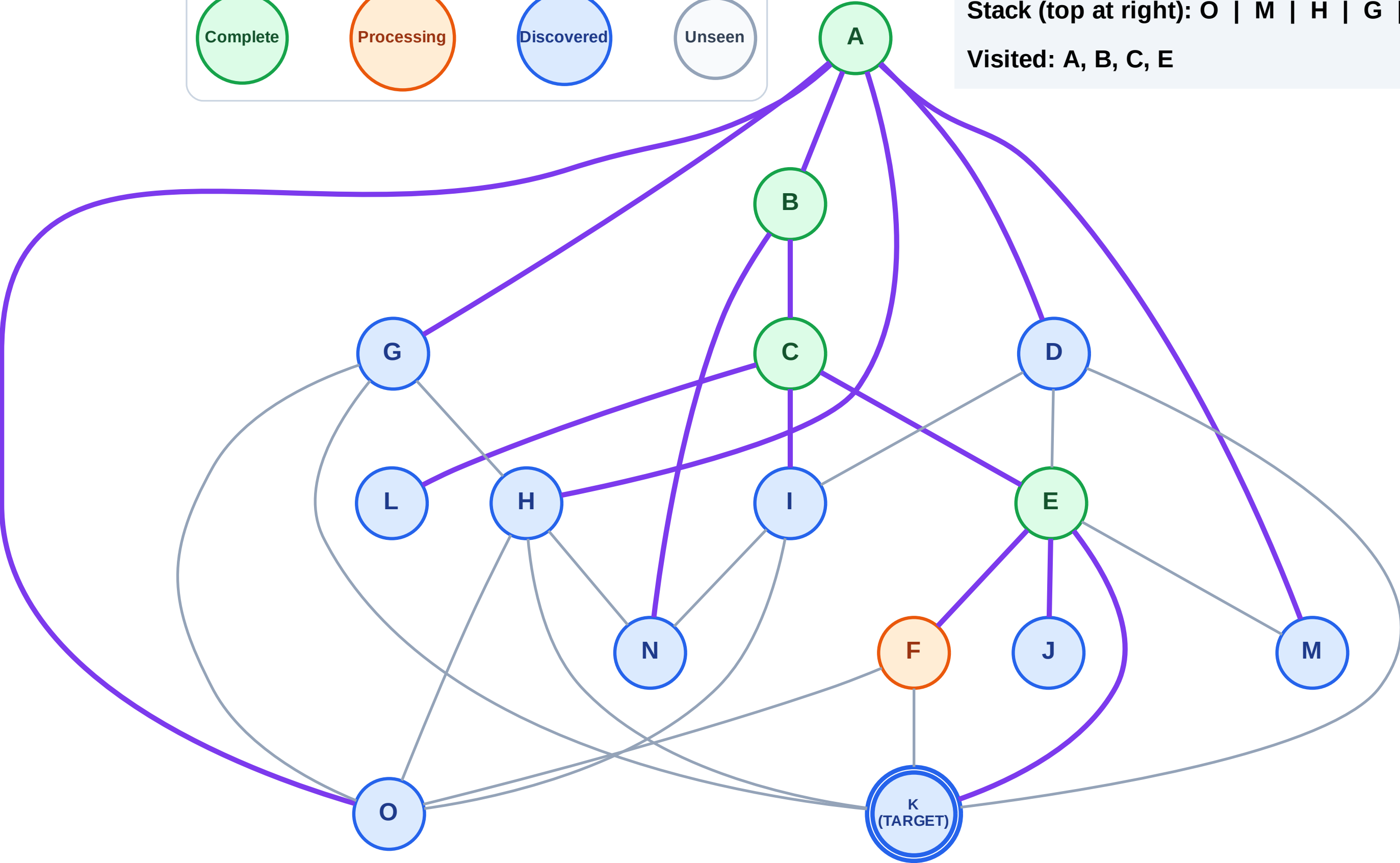
Processing

Discovered

Unseen

Stack (top at right): O | M | H | G | D | N | L | I | K | J

Visited: A, B, C, E



DFS Traversal

Step 11: No new neighbors from F

Legend

Complete

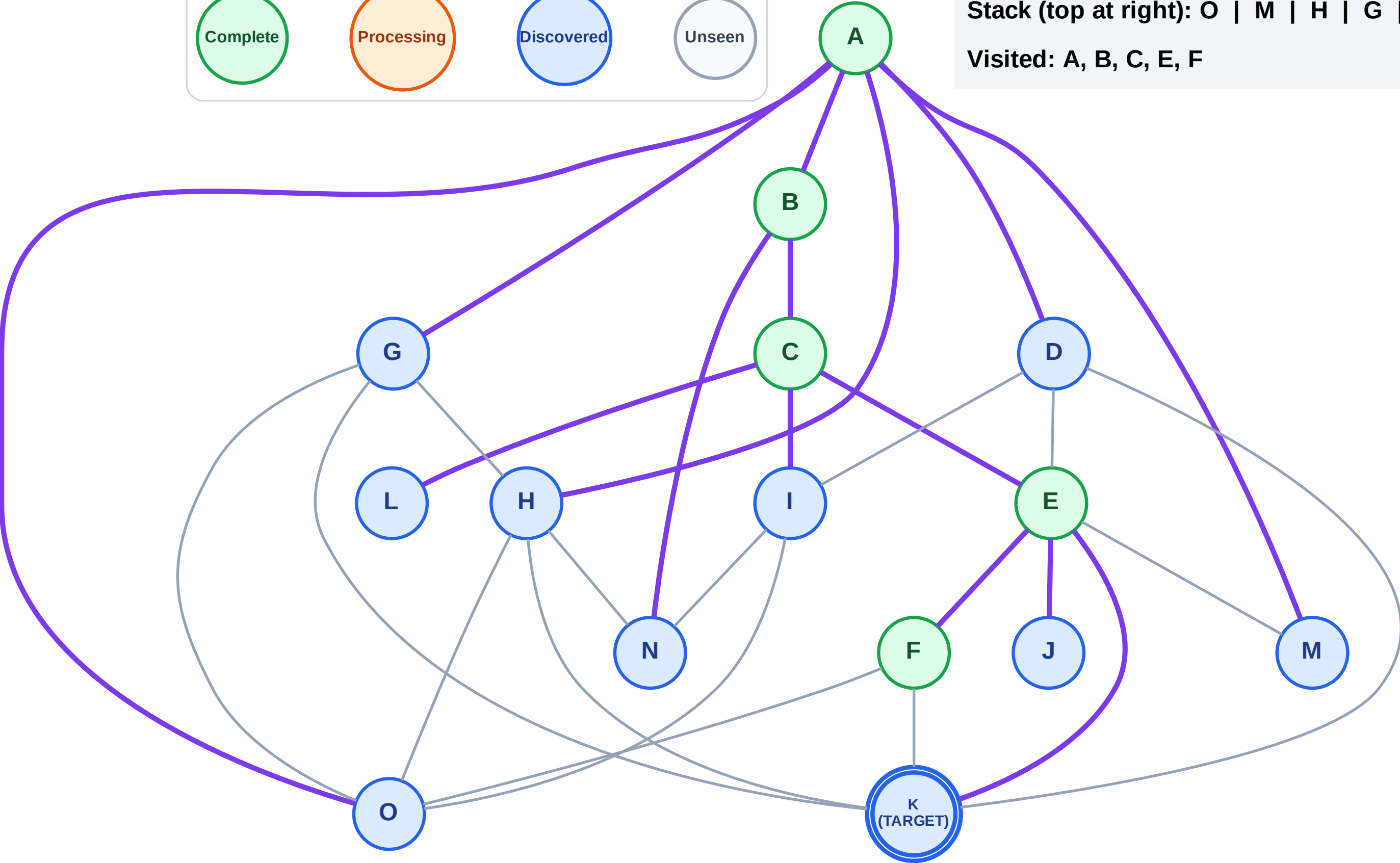
Processing

Discovered

Unseen

Stack (top at right): O | M | H | G | D | N | L | I | K | J

Visited: A, B, C, E, F



DFS Traversal

Step 12: Process node J

Legend

Complete

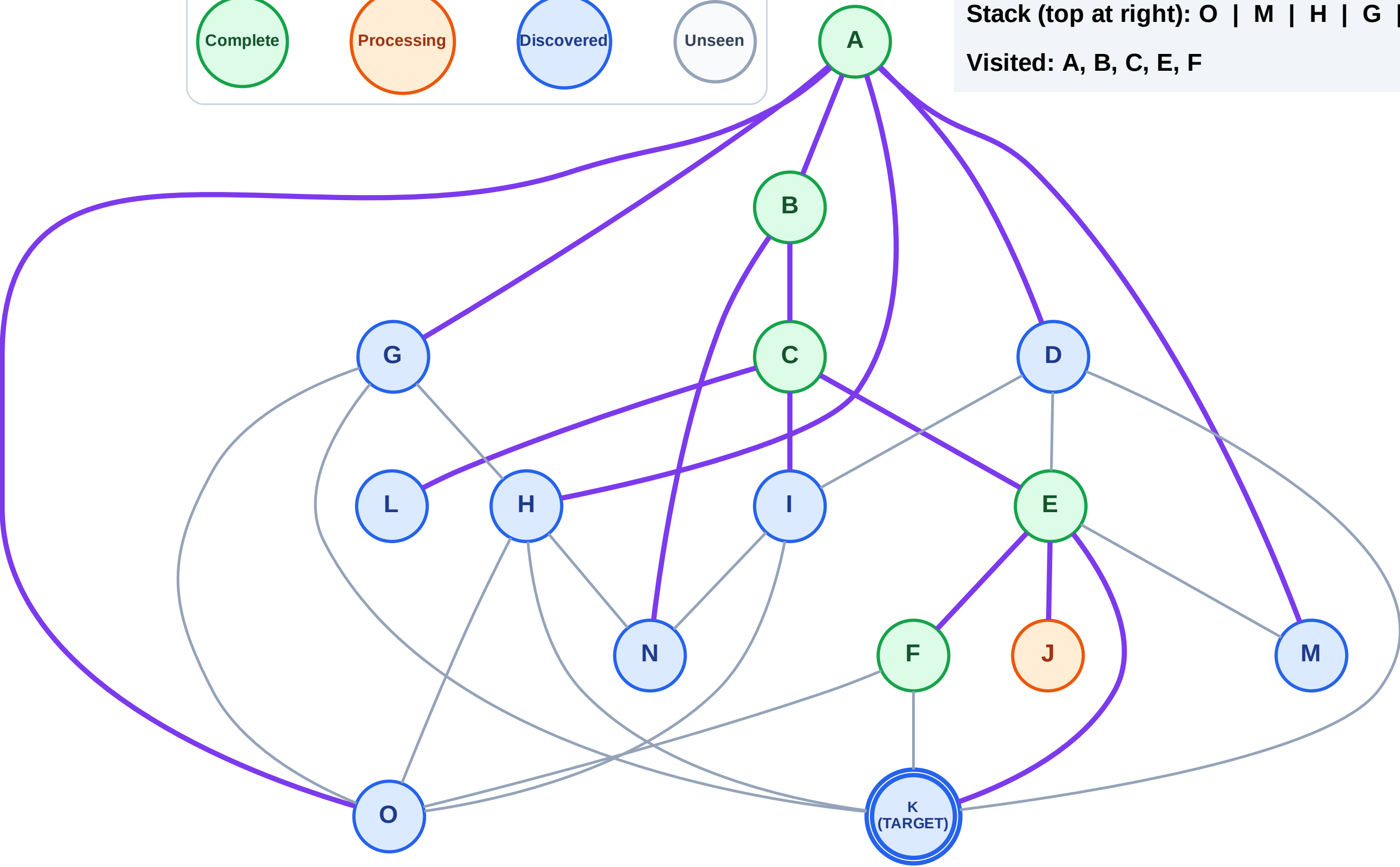
Processing

Discovered

Unseen

Stack (top at right): O | M | H | G | D | N | L | I | K

Visited: A, B, C, E, F



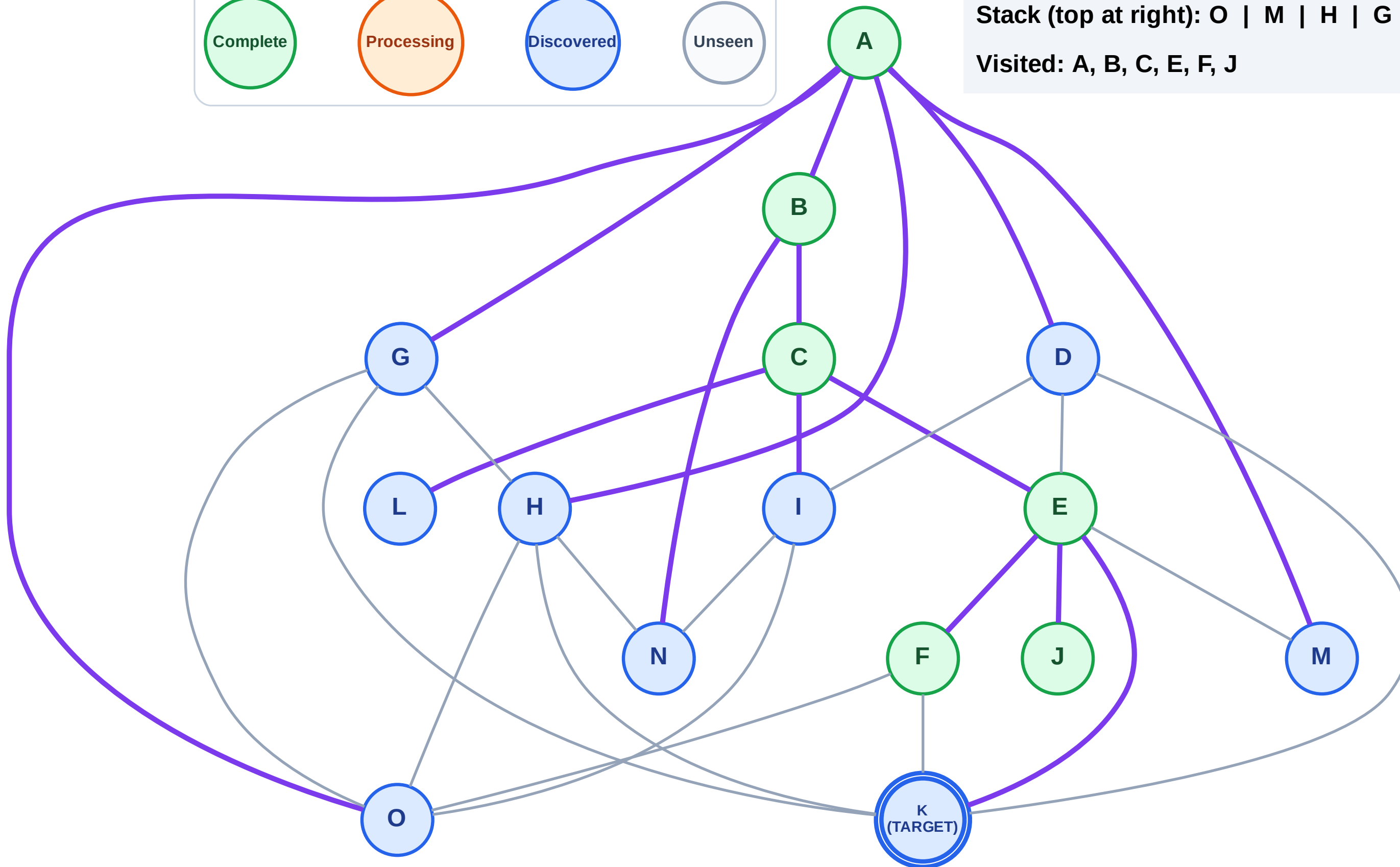
Step 13: No new neighbors from J

Legend



The legend consists of four colored circles arranged horizontally, each containing a label. From left to right: a green circle labeled 'Complete', an orange circle labeled 'Processing', a blue circle labeled 'Discovered', and a grey circle labeled 'Unseen'.

Visited: A, B, C, E, F, J



DFS Traversal

Step 14: Process node K

Legend

Complete

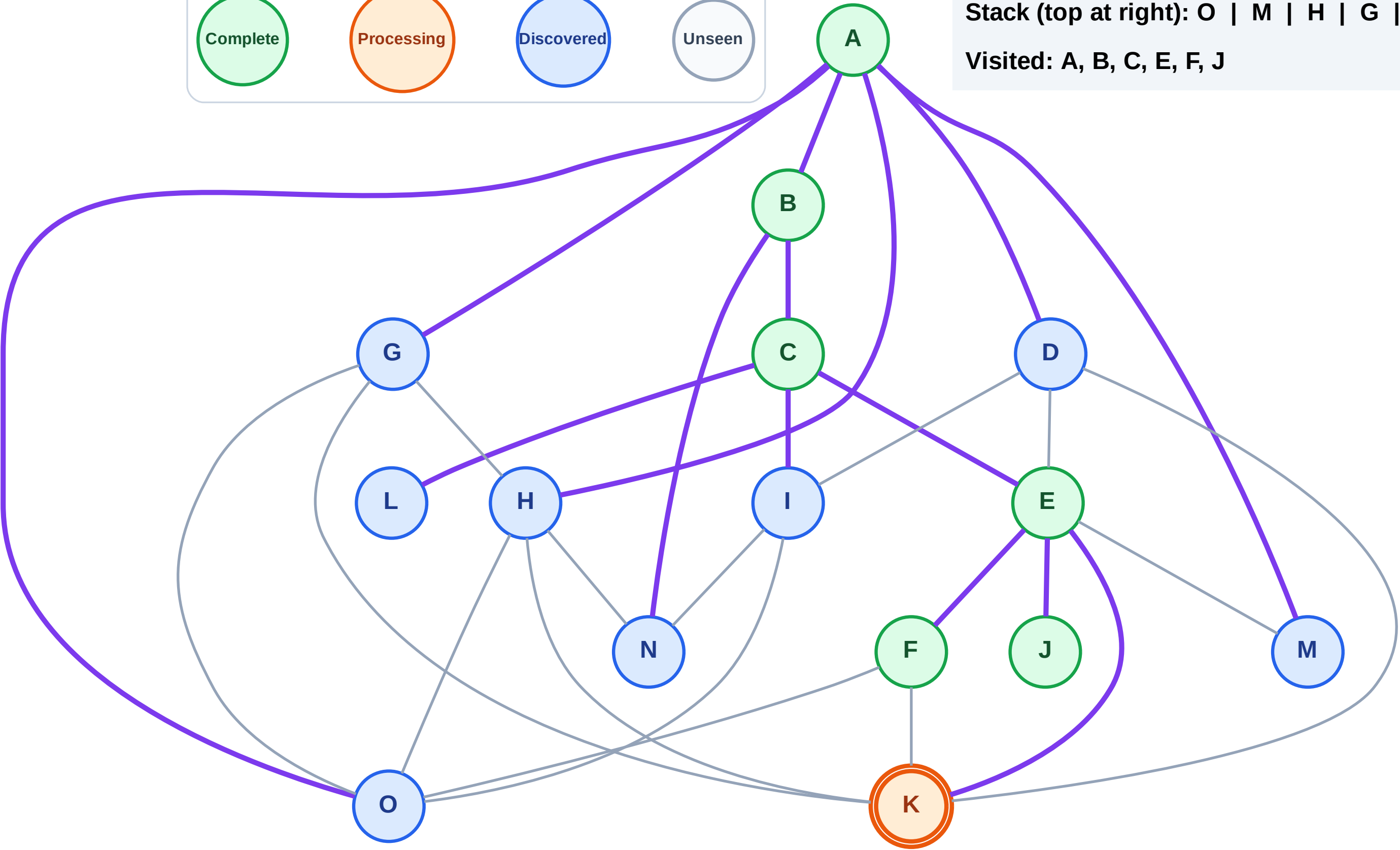
Processing

Discovered

Unseen

Stack (top at right): O | M | H | G | D | N | L | I

Visited: A, B, C, E, F, J



DFS Traversal

Step 15: TARGET NODE K REACHED! Path (4 hops): A → B → C → E → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | M | H | G | D | N | L | I

Visited: A, B, C, E, F, J, K

